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(54) **SALT-REJECTION SOLAR EVAPORATOR SYSTEM AND METHOD**

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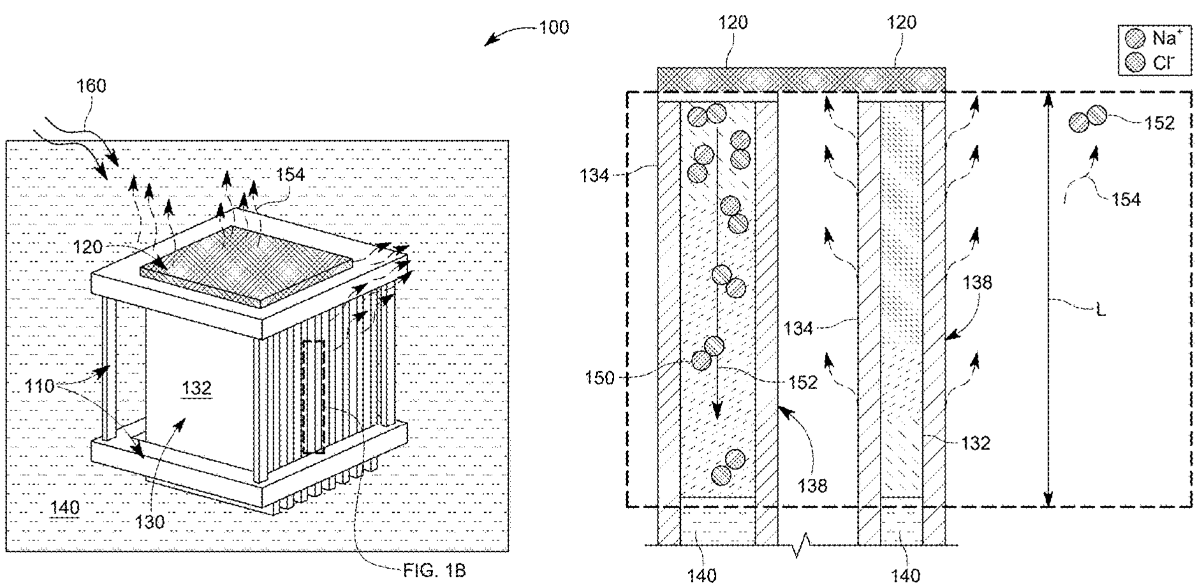
(57) **ABSTRACT**

A salt-rejection evaporator system includes a support frame, a mass and heat transport component supported by the support frame, the mass and heat transport component having plural transport layers, and a solar absorber layer located on top of the transport layers. The plural transport layers include plural microchannels that support capillarity, promote a flow of a saline feed toward the solar absorber layer and generate vapors due to heat generated by the solar absorber layer. The solar absorber layer is formed directly on top of the plural transport layers.



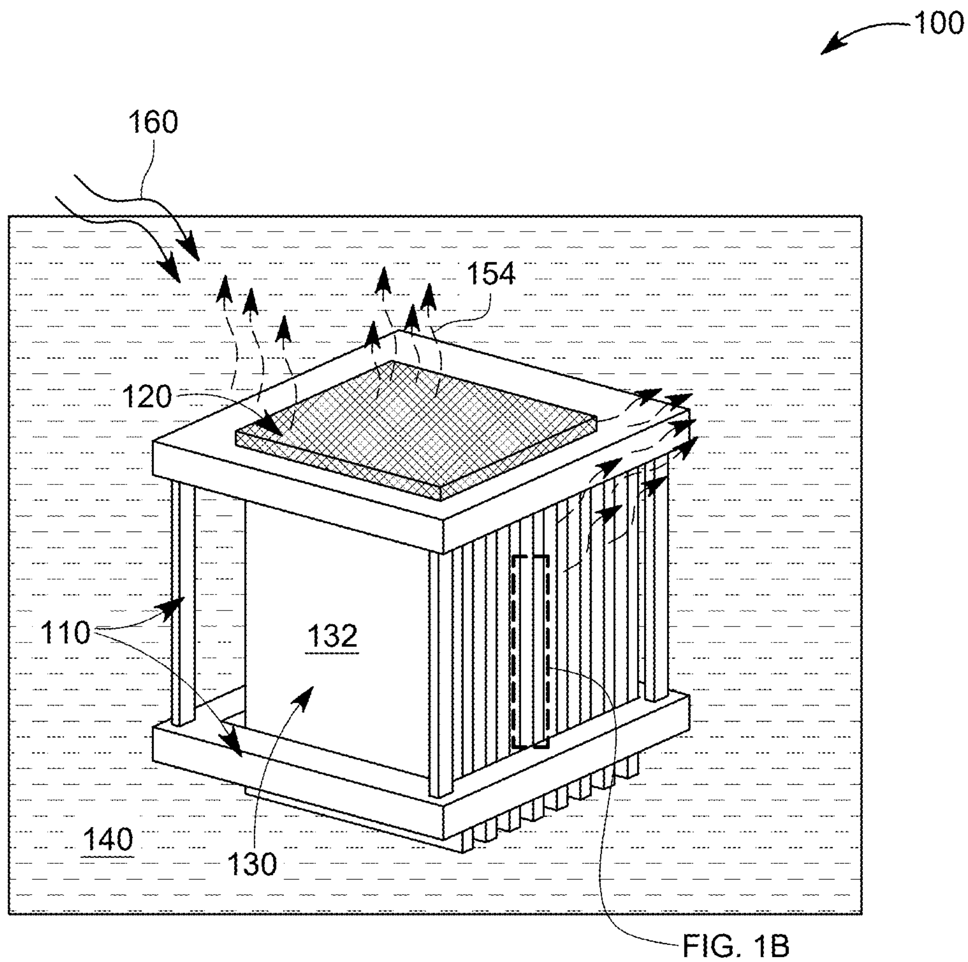


FIG. 1A

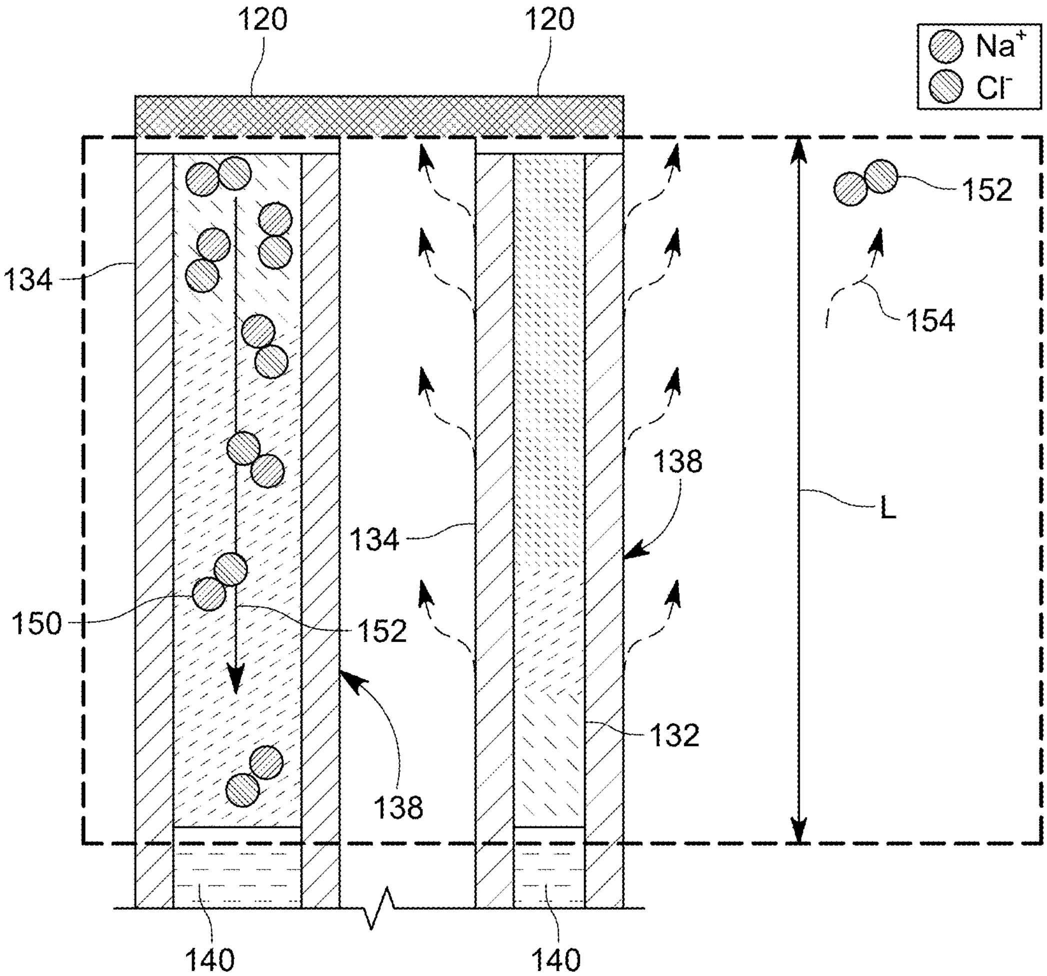


FIG. 1B

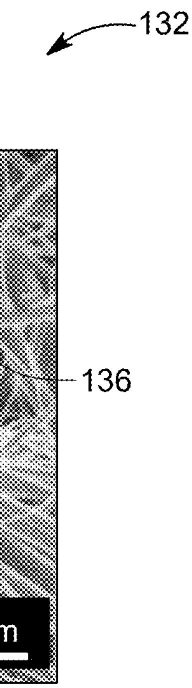


FIG. 2

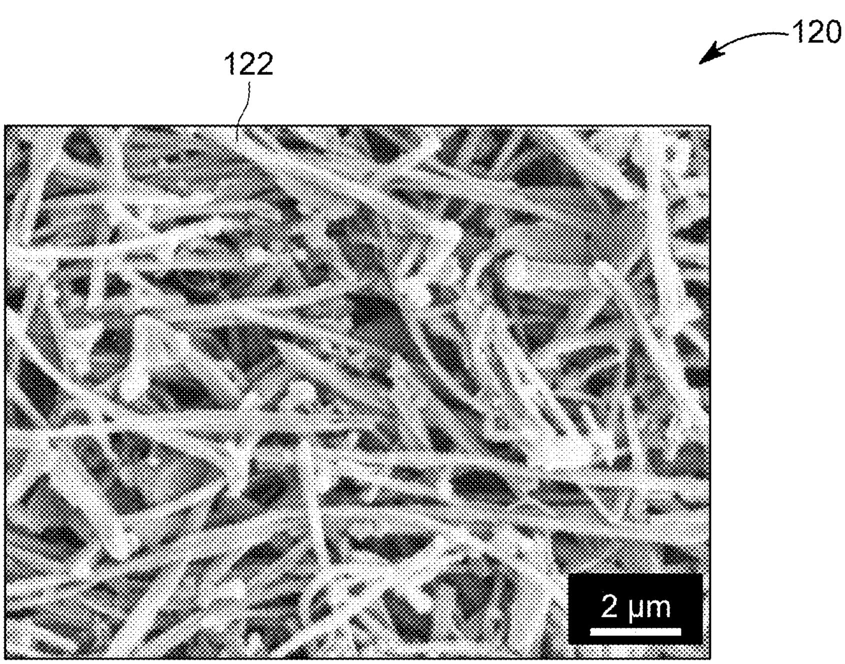


FIG. 3

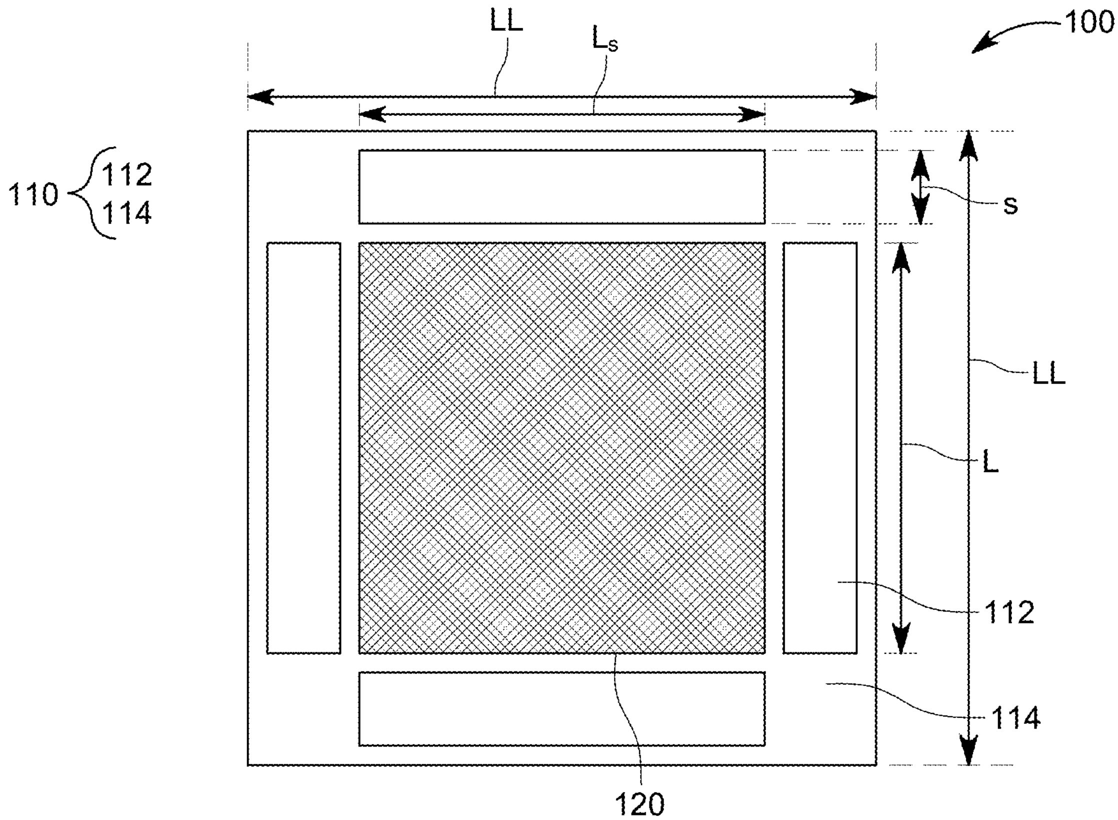

100
LL
L_s
110
112
114
s
LL
L
112
114
120

FIG. 4A

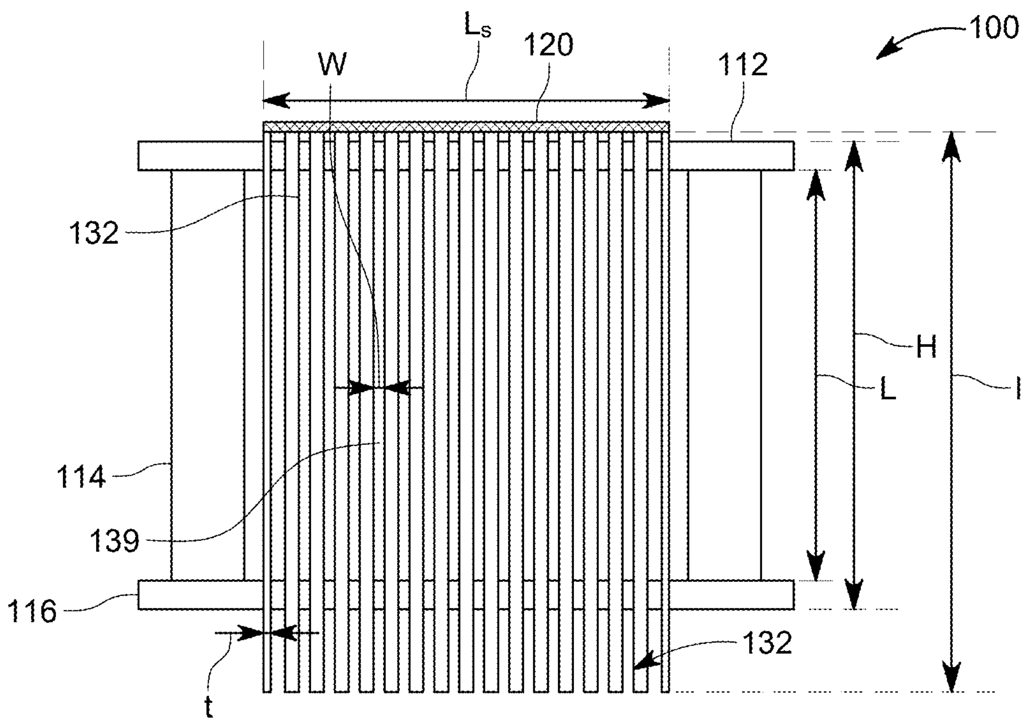

100
L_s
120
W
112
132
H
L
I
114
139
116
132
t

FIG. 4B

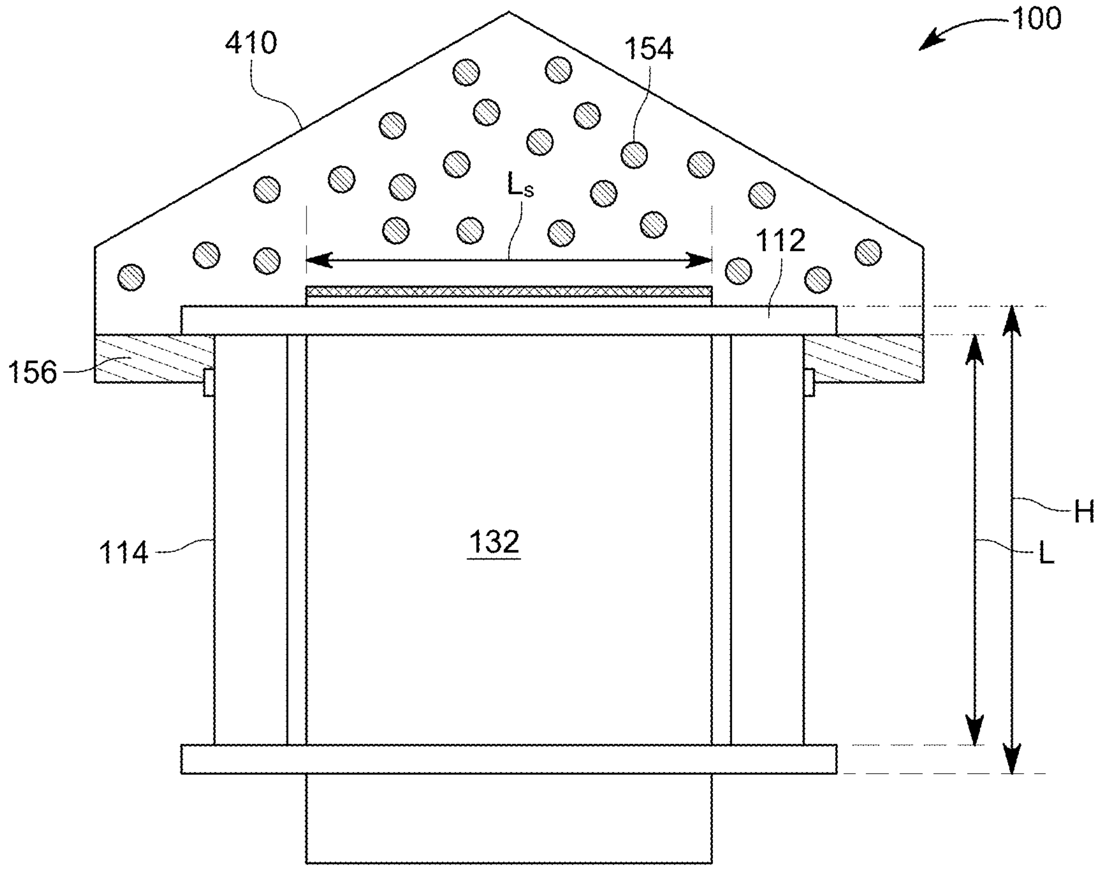


FIG. 4C

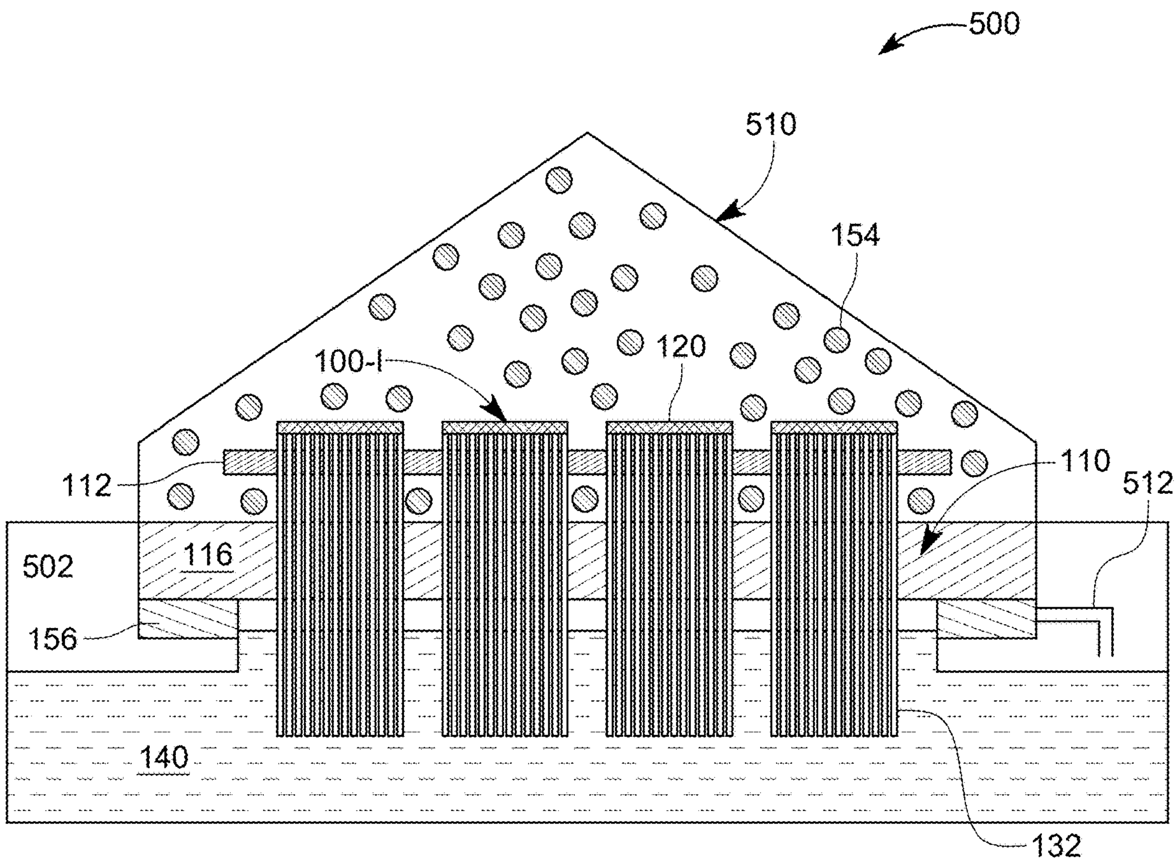


FIG. 5

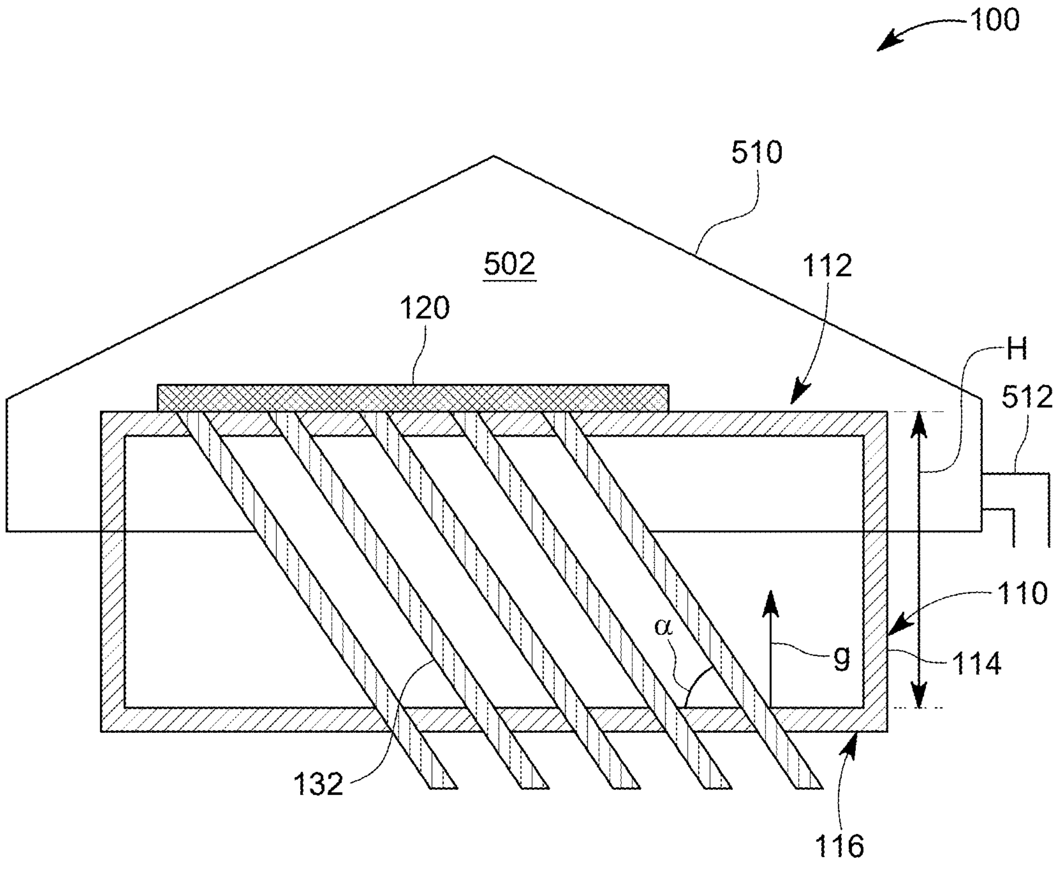


FIG. 6A

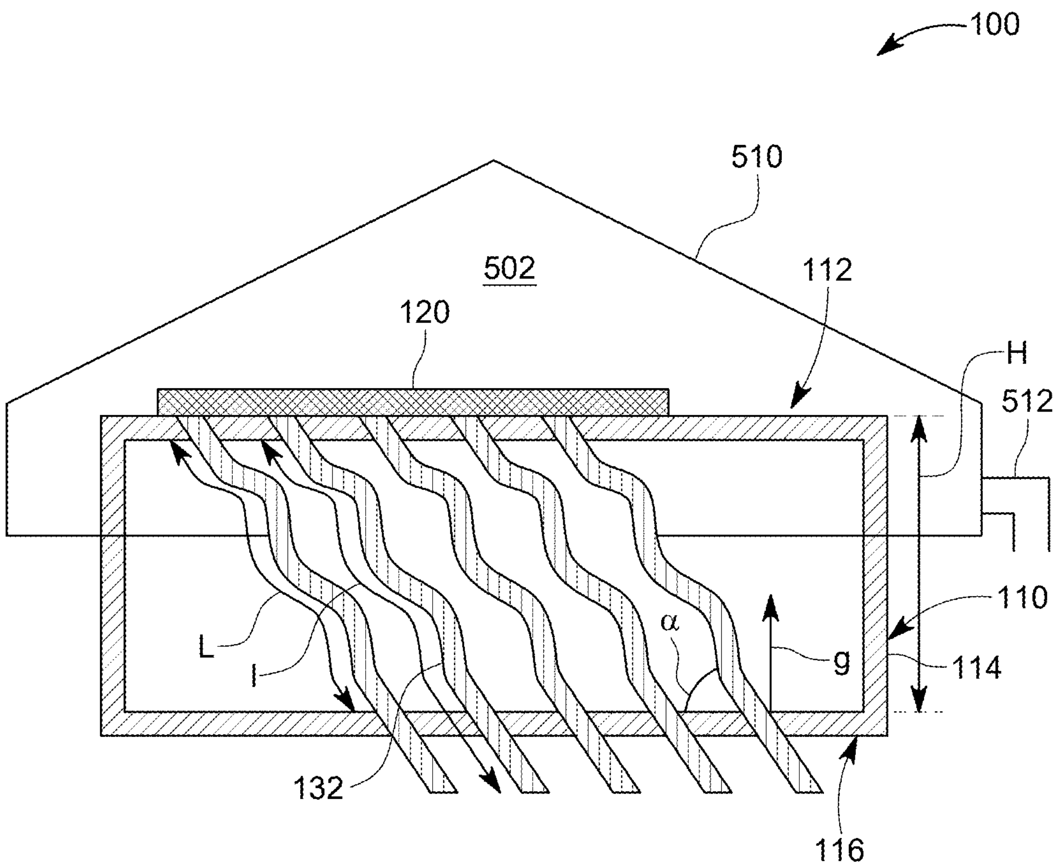


FIG. 6B

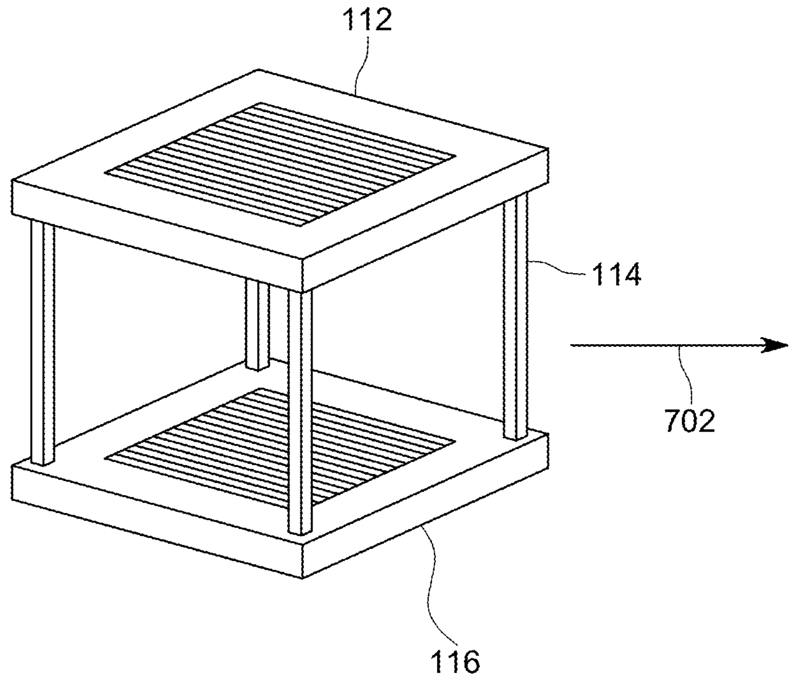


FIG. 7A

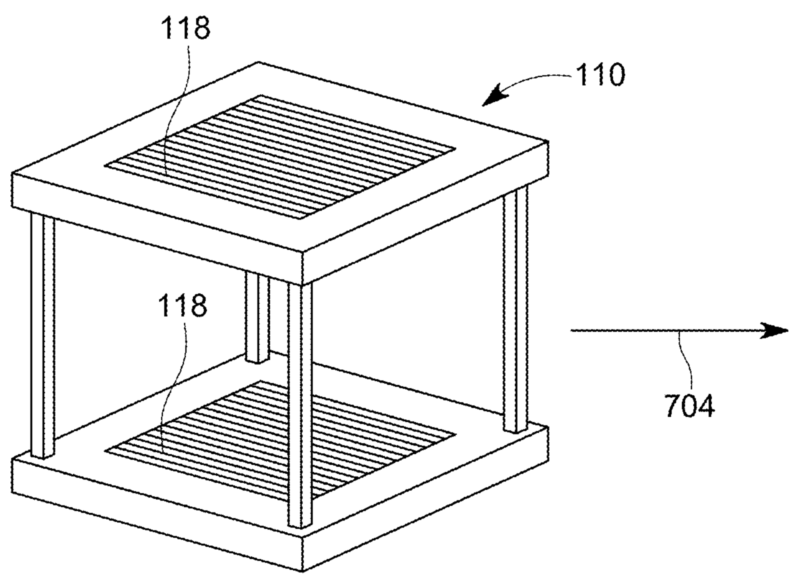


FIG. 7B

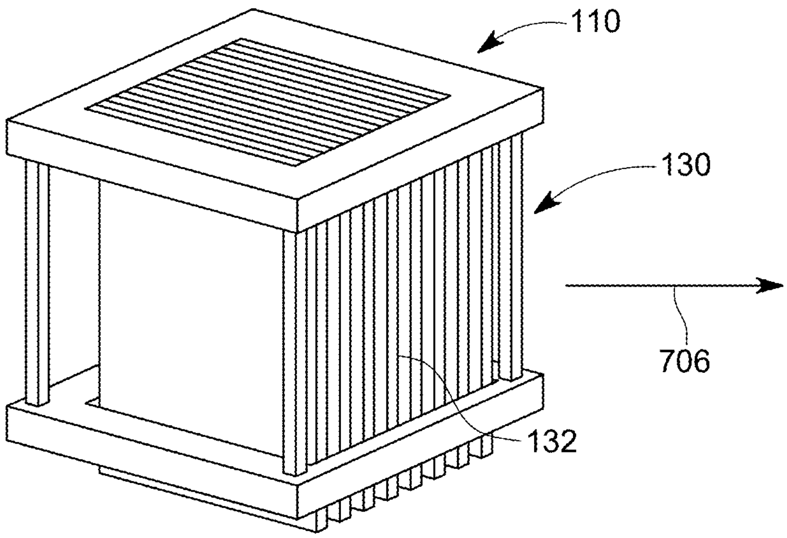


FIG. 7C

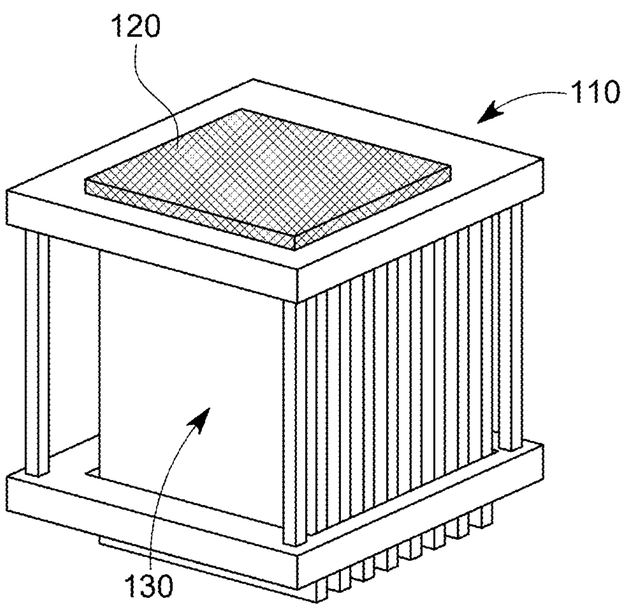


FIG. 7D

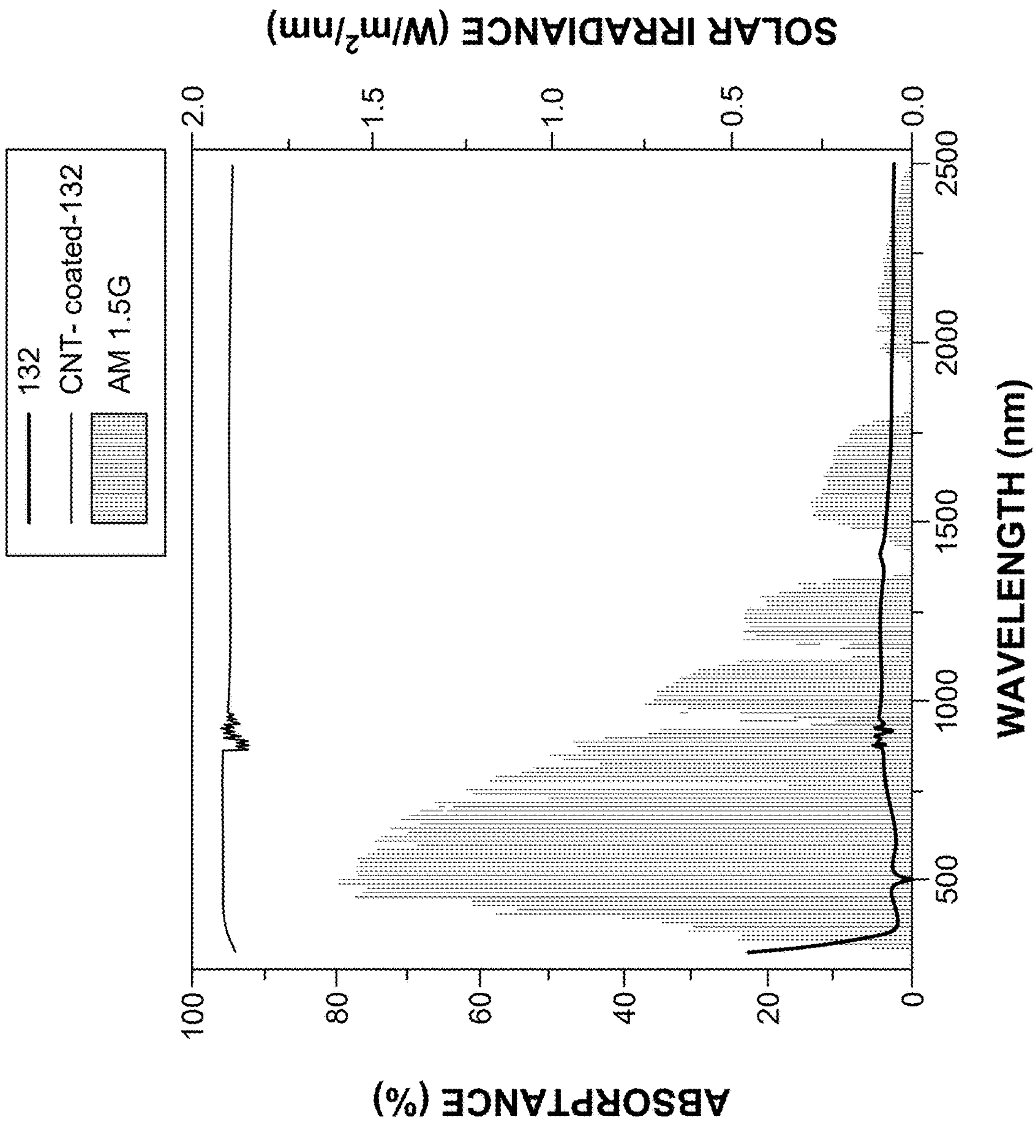

SOLAR IRRADIANCE (W/m^2/nm)
2.0
1.5
1.0
0.5
0.0
132
CNT- coated-132
AM 1.5G
100
80
60
40
20
0
ABSORPTANCE (%)
500
1000
1500
2000
2500
WAVELENGTH (nm)

FIG. 8

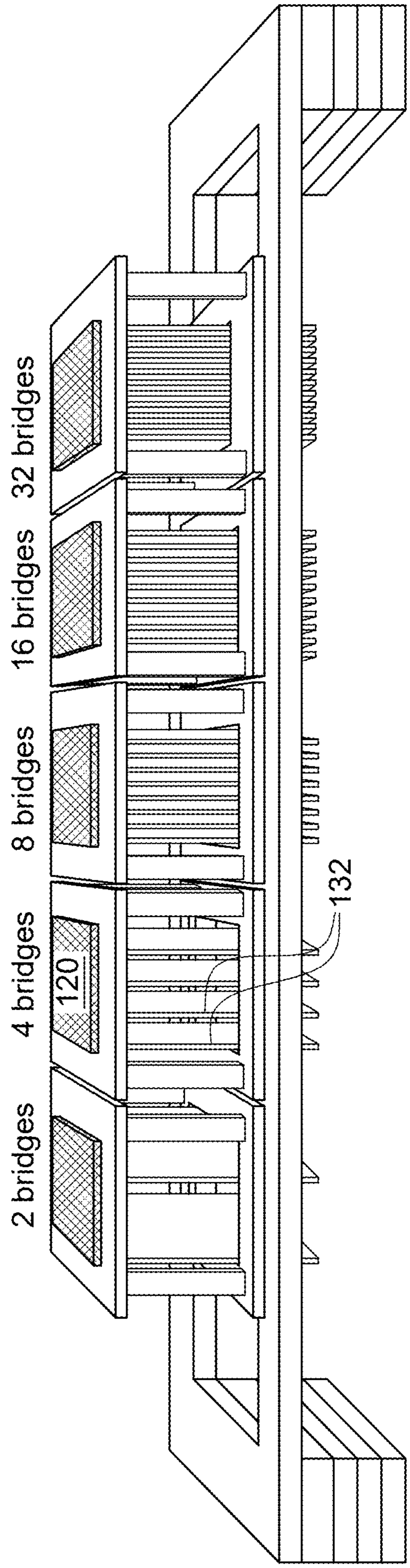


FIG. 9

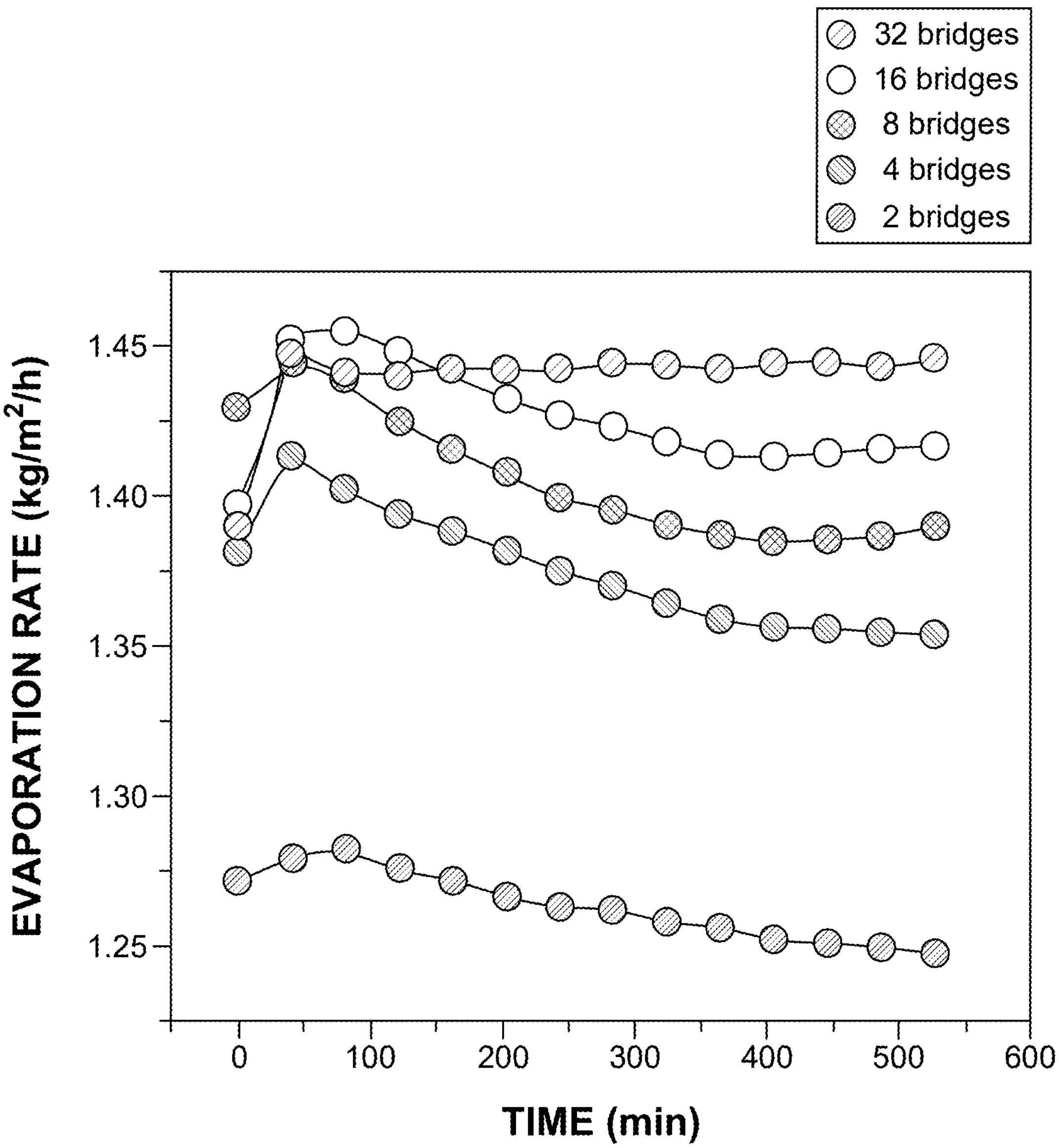

32 bridges
16 bridges
8 bridges
4 bridges
2 bridges
EVAPORATION RATE (kg/m2/h)
1.45
1.40
1.35
1.30
1.25
0
100
200
300
400
500
600
TIME (min)

FIG. 10

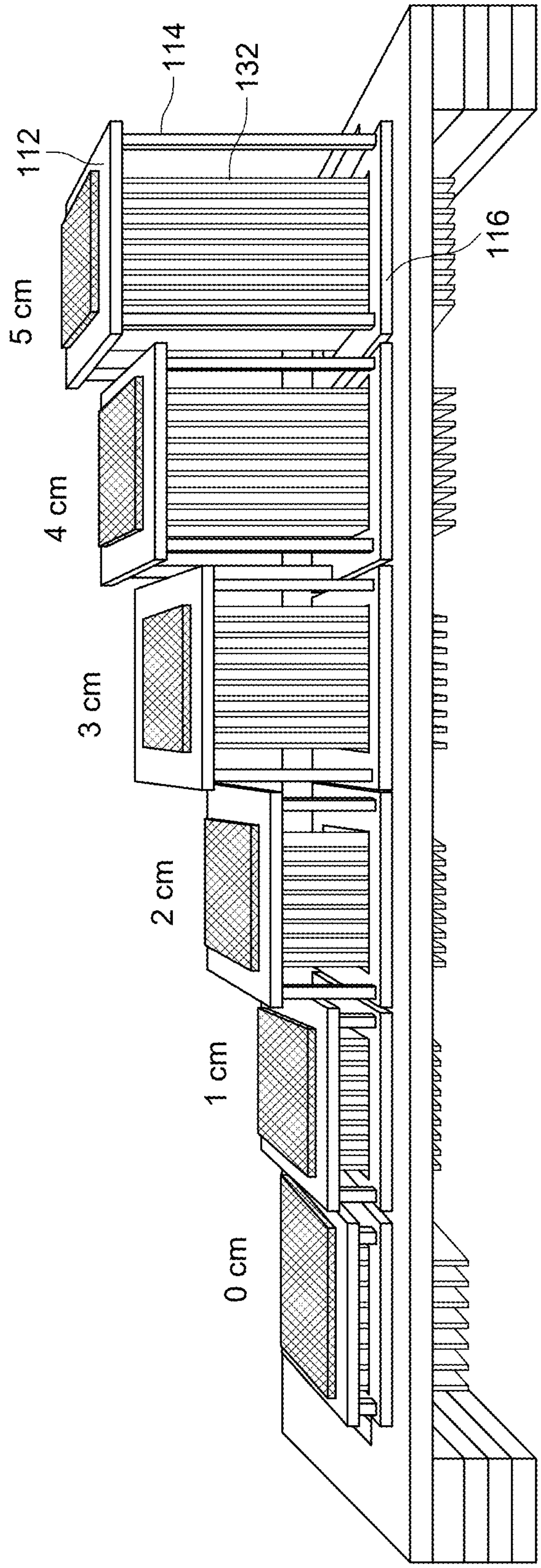


FIG. 11

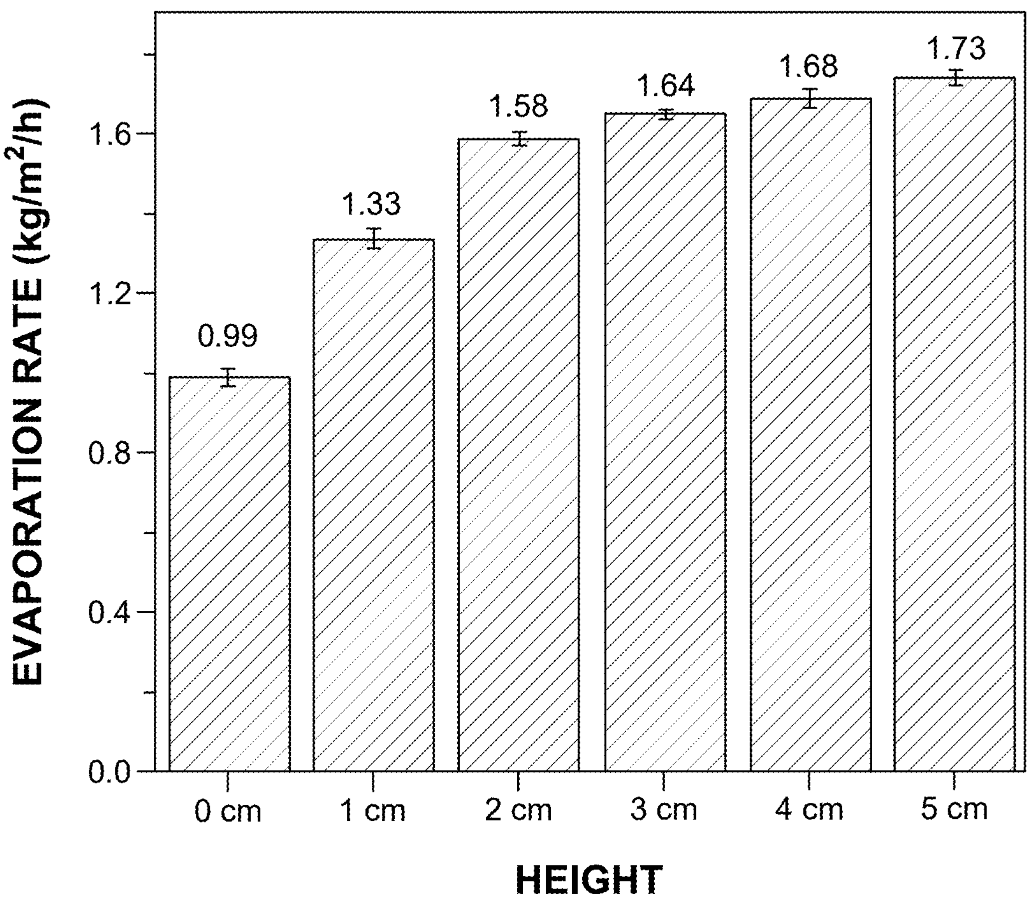

EVAPORATION RATE (kg/m²/h)
0.0
0.4
0.8
1.2
1.6
0.99
1.33
1.58
1.64
1.68
1.73
0 cm
1 cm
2 cm
3 cm
4 cm
5 cm
HEIGHT

FIG. 12

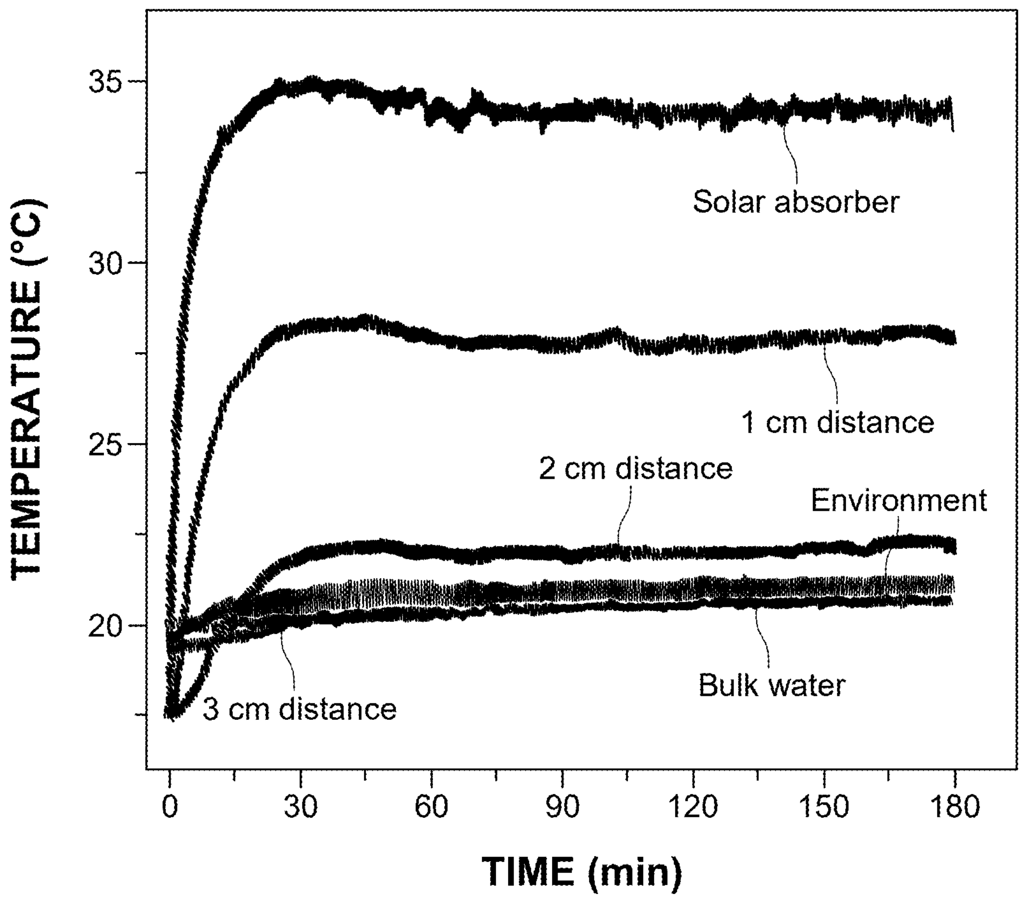

35
30
25
20
TEMPERATURE (°C)
0
30
60
90
120
150
180
TIME (min)
Solar absorber
1 cm distance
2 cm distance
Environment
Bulk water
3 cm distance

FIG. 13

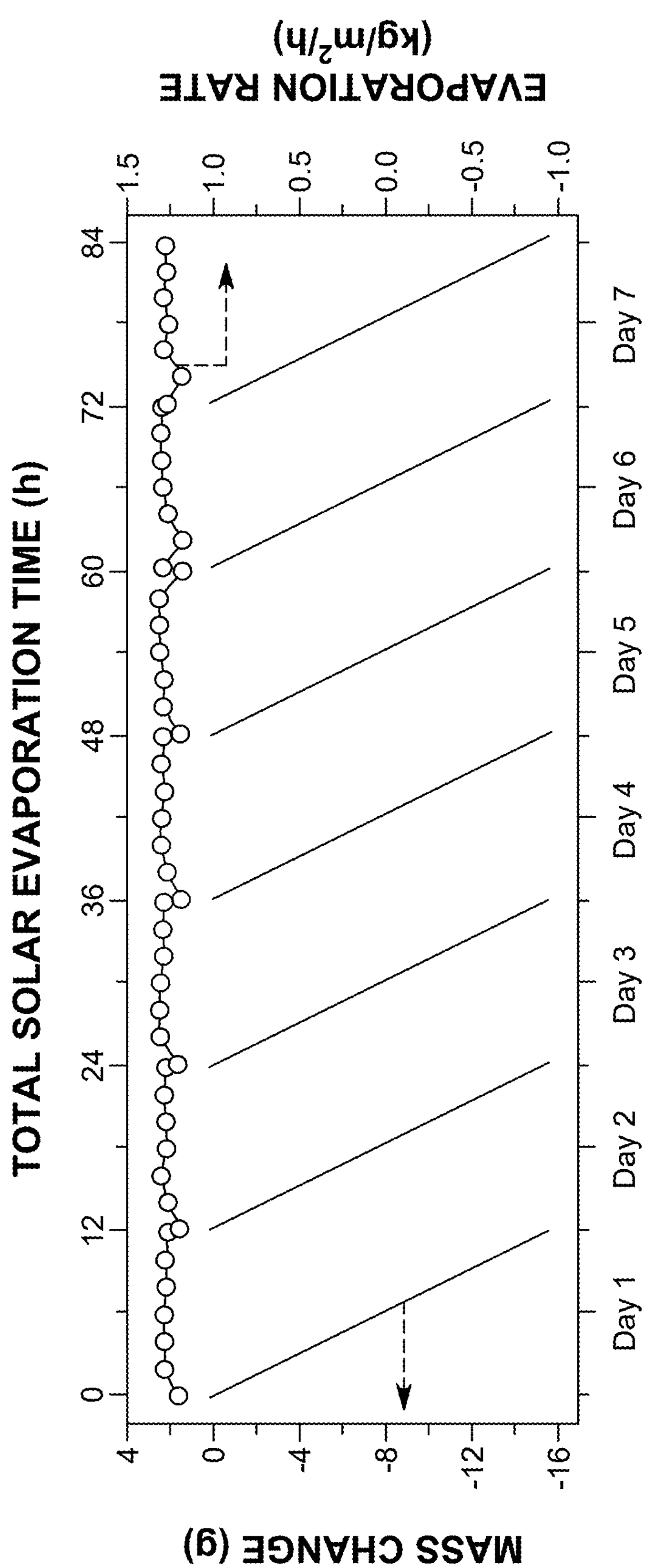

TOTAL SOLAR EVAPORATION TIME (h)
0
12
24
36
48
60
72
84
EVAPORATION RATE
(kg/m2/h)
1.5
1.0
0.5
0.0
-0.5
-1.0
MASS CHANGE (g)
4
0
-4
-8
-12
-16
Day 1
Day 2
Day 3
Day 4
Day 5
Day 6
Day 7

FIG. 14

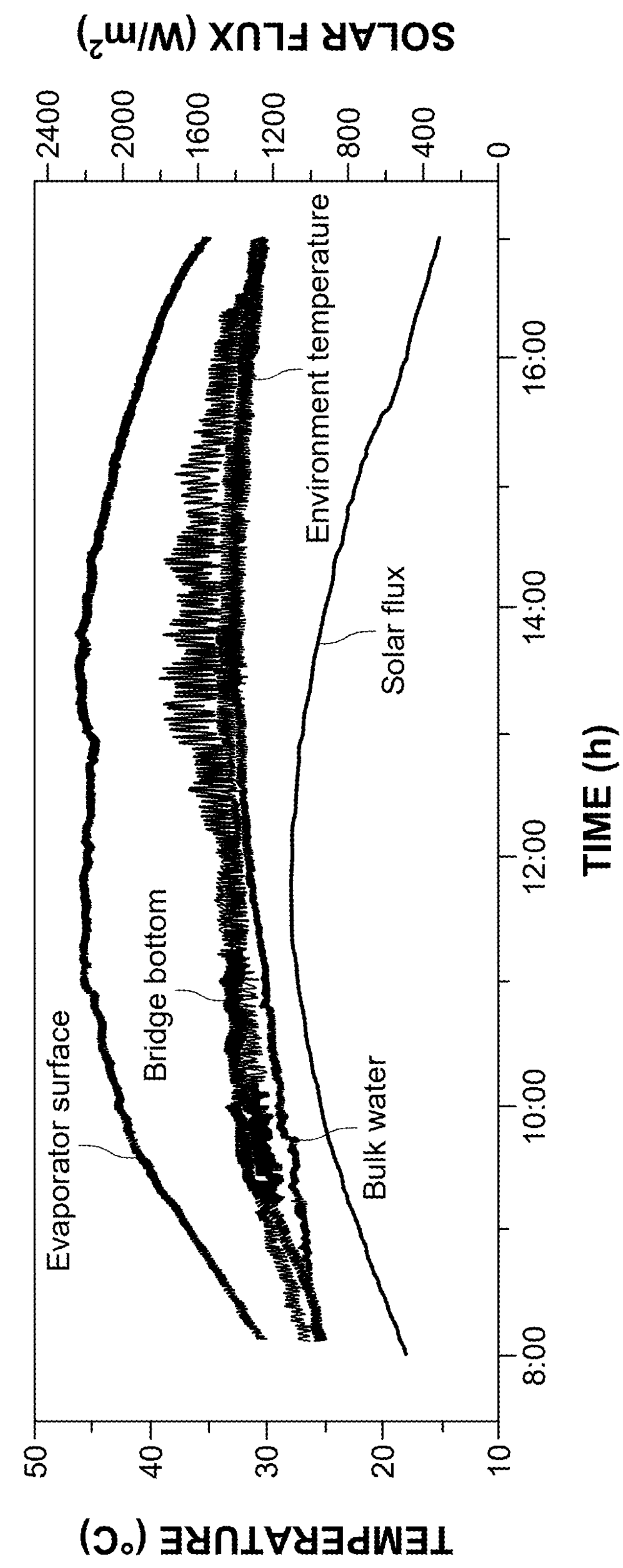

SOLAR FLUX (W/m²)
2400
2000
1600
1200
800
400
0
TEMPERATURE (°C)
50
40
30
20
10
TIME (h)
8:00
10:00
12:00
14:00
16:00
Evaporator surface
Bridge bottom
Bulk water
Solar flux
Environment temperature

FIG. 15

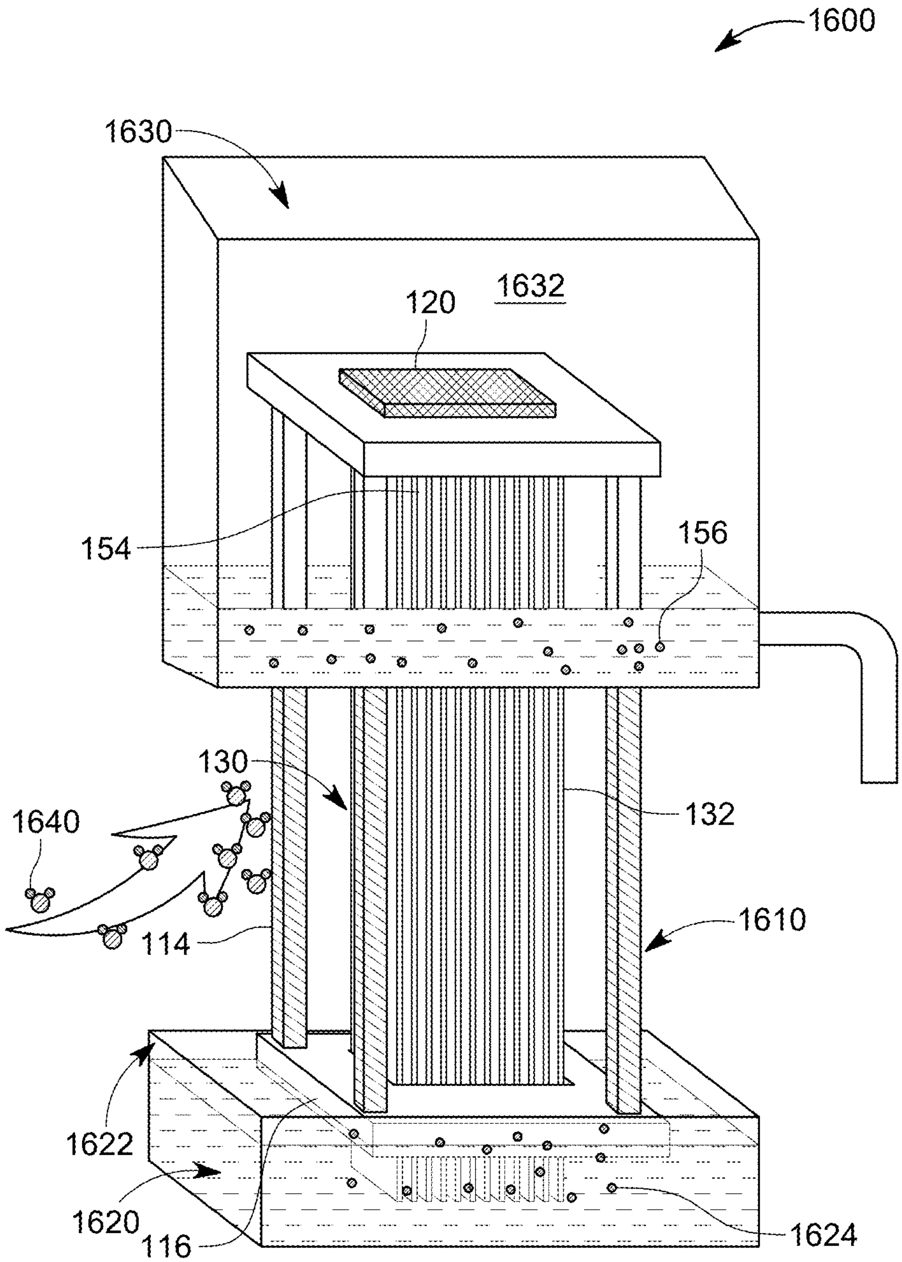

1600
1630
1632
120
154
156
130
132
1640
114
1610
1622
1620
116
1624

FIG. 16

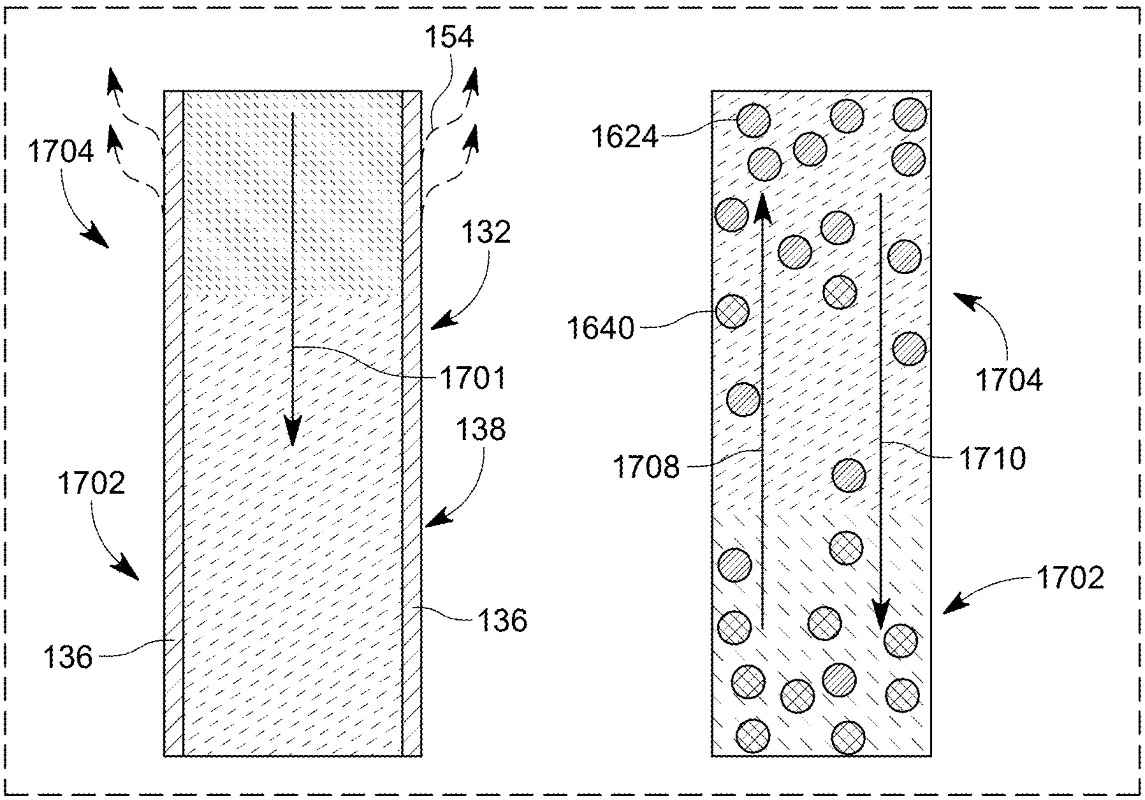

154
1704
132
1624
1640
1701
1704
138
1708
1710
1702
1702
136
136

FIG. 17

SALT-REJECTION SOLAR EVAPORATOR SYSTEM AND METHOD

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to U.S. Provisional Patent Application No. 63/342,781, filed on May 17, 2022, entitled "VERTICALLY ALIGNED MASS TRANSPORT BRIDGES WITH MINIMIZED CONDUCTIVE HEAT LOSS FOR SELF-CLEANING SOLAR DESALINATION," the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND OF THE INVENTION

Technical Field

[0002] Embodiments of the subject matter disclosed herein generally relate to a system and method for direct solar desalination, and more particularly, to a three-dimensional, open architecture for such a system that enables salt-rejection and increased water production efficiency due to plural transport layers.

Discussion of the Background

[0003] Freshwater scarcity is a paramount global challenge, impacting 2.2 billion people, particularly those residing in arid and remote regions, where transporting water over long distances is either costly or impractical. This situation has been exacerbated by climatic change and rapid population growth. Agricultural irrigation and electrical power generation are the two primary consumers of freshwater, accounting for 70% and 15% of global freshwater withdrawals, respectively. However, freshwater production is reliant on energy input, underscoring the critical importance of the water-energy-food *nexus* in achieving sustainable development. The production of freshwater for drinking and irrigation using renewable energy sources is crucial in mitigating the mounting pressure resulting from escalating demands for water, energy, and food.

[0004] As a sustainable desalination technology, direct solar desalination has the advantages of low cost, off-grid capability, and zero carbon footprint, and is particularly useful for remote areas and distributed communities. The solar-to-vapor conversion efficiency for single-stage processes has approached 100% in the past few years. The evaporation rate can even surpass the theoretical limit for exclusive solar-driven evaporation by exploiting the environmental heat. However, high evaporation rates cannot be maintained during the evaporation of saline water (e.g., seawater and concentrated brine discharged from reverse osmosis (RO) facilities) because of the salt accumulation in the system. For instance, commercial solar stills (e.g., Aquamate Solar Still®) cannot work for practical desalination applications because their evaporators cannot be replaced or cleaned, thus resulting in a short lifespan [1].

[0005] Recently, innovative strategies, which can be classified into two general categories, (1) hydrophobic light-absorbing layer design and (2) fluid convection enhancement, were developed to address these challenges. For example, the Janus structures having an upper hydrophobic solar-absorber layer and an underlying hydrophilic water-absorbing layer were proposed [2]. In this design, the saline water cannot reach the upper surface because of its hydrophobicity, thereby preventing surface salt accumulation. However, despite the excellent salt rejection capability associated with this design, its energy conversion efficiency is limited by the rapid heat dissipation from the solar absorber to the bulk water underneath. Likewise, although salt rejection can be realized by improving the fluid convection between the evaporation surface and bulk water, the fluid exchange not only removes salt but also takes heat away from the evaporation surface, thereby resulting in a relatively low vapor generation rate [3].

[0006] Thus, the paradox between salt rejection and heat loss remains one of the most challenging barriers faced by solar-driven interfacial evaporation strategies and for this reason, there is a need for a new system that is capable of generating high efficiency water production while also achieving salt-rejection so that the new device can operate for a long term without the need to change any of its parts and also without the need to provide external power.

SUMMARY OF THE INVENTION

[0007] According to an embodiment, there is a salt-rejection evaporator system that includes a support frame, a mass and heat transport component supported by the support frame, the mass and heat transport component having plural transport layers, and a solar absorber layer located on top of the transport layers. The plural transport layers include plural microchannels that support capillarity, promote a flow of a saline feed toward the solar absorber layer and generate vapors due to heat generated by the solar absorber layer. The solar absorber layer is formed directly on top of the plural transport layers.

[0008] According to another embodiment, there is a solar-driven atmospheric water extraction system that includes a support frame, a mass and heat transport component supported by the support frame, the mass and heat transport component having plural transport layers, a solar absorber layer located on top of the transport layers, and a sorption system in which a bottom of the plural transport layers is located. The plural transport layers include plural microchannels that support capillarity, promote a flow of atmospheric water toward the solar absorber layer, generate vapors due to heat generated by the solar absorber layer, and absorb atmospheric water and the solar absorber layer is formed directly on top of the plural transport layers.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] For a more complete understanding of the present invention, reference is now made to the following descriptions taken in conjunction with the accompanying drawings, in which:

[0010] FIG. **1**A is a schematic diagram of a salt-rejection evaporator system having plural transport layers extending upwards, from a saline feed to a solar absorber layer;

[0011] FIG. **1**B schematically illustrates a mass transport and heat transport through transport paths formed within the transport layers of the salt-rejection evaporator system;

[0012] FIG. **2** schematically illustrates the structure of a transport layer of the salt-rejection evaporator system and the transport paths formed inside the transport layer;

[0013] FIG. **3** schematically illustrates the structure of a solar absorber layer used by the salt-rejection evaporator system to heat the transported saline feed;

[0014] FIGS. **4**A to **4**C show various views of the salt-rejection evaporator system;
[0015] FIG. **5** is a schematic diagram of a salt-rejection evaporator system having plural individual salt-rejection evaporator systems connected to a single support frame;
[0016] FIGS. **6**A and **6**B illustrate various shapes and positions of the transport layers of the salt-rejection evaporator system;
[0017] FIGS. **7**A to **7**D illustrate a method for making the salt-rejection evaporator system of FIG. **1**;
[0018] FIG. **8** illustrates the ultraviolet, visible, near-infrared (UV-vis-NIR) spectra of the transport layer, the solar absorber layer coated on the transport layers, and a standard irradiation spectrum of AM 1.5 G;
[0019] FIG. **9** illustrates salt-rejection evaporator systems having various numbers of transport layers;
[0020] FIG. **10** illustrates the evaporation rate corresponding to the salt-rejection evaporator systems of FIG. **9**;
[0021] FIG. **11** shows various height salt-rejection evaporator systems;
[0022] FIG. **12** shows the evaporation rate for different heights of the transport layers in the salt-rejection evaporator system;
[0023] FIG. **13** shows the temperature distribution inside the transport layers of the systems of FIG. **11**;
[0024] FIG. **14** illustrates the mass change curves and evaporation rates of the evaporator system during a time interval of a few days;
[0025] FIG. **15** illustrates a real-time temperature variation of the solar absorber layer, environment, bottom of the transport layers, and the saline feed for a test performed over a few days in the real environment;
[0026] FIG. **16** illustrates a sorption-based atmospheric water extraction system using plural transport layers; and
[0027] FIG. **17** schematically illustrates the temperature gradient through the microchannels of the plural transport layers, and the back flow of the sorption particles.

DETAILED DESCRIPTION OF THE INVENTION

[0028] The following description of the embodiments refers to the accompanying drawings. The same reference numbers in different drawings identify the same or similar elements. The following detailed description does not limit the invention. Instead, the scope of the invention is defined by the appended claims. The following embodiments are discussed, for simplicity, with regard to a salt-rejection solar evaporator system that has an open architecture and includes plural vertical mass transport layers. However, the embodiments to be discussed next are not limited to vertical mass transport layers, but may be applied to non-vertical mass transport layers.

[0029] Reference throughout the specification to "one embodiment" or "an embodiment" means that a particular feature, structure or characteristic described in connection with an embodiment is included in at least one embodiment of the subject matter disclosed. Thus, the appearance of the phrases "in one embodiment" or "in an embodiment" in various places throughout the specification is not necessarily referring to the same embodiment. Further, the particular features, structures or characteristics may be combined in any suitable manner in one or more embodiments.

[0030] It will be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first object or step could be termed a second object or step, and, similarly, a second object or step could be termed a first object or step, without departing from the scope of the present disclosure. The first object or step, and the second object or step, are both, objects or steps, respectively, but they are not to be considered the same object or step.

[0031] The terminology used in the description herein is for the purpose of describing particular embodiments and is not intended to be limiting. As used in this description and the appended claims, the singular forms "a," "an" and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term "and/or" as used herein refers to and encompasses any possible combinations of one or more of the associated listed items. It will be further understood that the terms "includes," "including," "comprises" and/or "comprising," when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. Further, as used herein, the term "if" may be construed to mean "when" or "upon" or "in response to determining" or "in response to detecting," depending on the context.

[0032] According to an embodiment, a novel three-dimensional (3D) salt-rejection evaporator system is presented and this new evaporator system achieves stable and efficient water evaporation by enhancing the salt backflow and conductive heat recovery. The evaporator system includes a number of vertically (or non-vertically) aligned mass transfer bridges (MTBs) or simply "transport layers" containing abundant hydrophilic microchannels. In addition to facilitating salt and water transport, the transport layers separate a solar absorber layer from the bulk water (saline feed), thus creating a highly open 3D space through which additional water can be evaporated by conductive heat from the solar absorber layer. The solar-driven vapor generation rate and practical water production performance of this 3D evaporator system were evaluated under laboratory and outdoor conditions and the experimental results demonstrate that the 3D evaporator system can stably and continuously operate without salt accumulation when processing high-salinity water (e.g., 12-14 wt. % NaCl solution and concentrated brine from RO facilities) while achieving a high vapor generation rate of about 1.64 $kg/m^2/h$. A scaled-up solar evaporator system was tested on the top of a building to demonstrate its practical applications. Moreover, a daily water collection rate of about 5 L/m^2 with a practical solar-water collection efficiency of >40% was demonstrated, better than a previously reported record of about 22% [4]. The 3D solar evaporator system is now discussed herein in more detail with regard to the figures.

[0033] FIG. **1**A shows a salt-rejection evaporator system **100** (note that other systems are discussed later) that includes a support frame **110**, a solar absorber layer **120** that is supported at the top of the evaporator system, by the support frame **110**, and a mass and heat transport component **130** (having plural layer **132**). System **100** is configured to be placed on a saline feed **140**, for example, sea water, industrial brine, wastewater, etc., which has a certain amount of salt. System **100** may be configured, as discussed later, to float on the saline feed. In one application, system **100** is fixedly placed at a certain location and saline feed **140** is supplied to the system.

[0034] The mass and heat transport component **130** includes, in this embodiment, plural transport layers or bridges or membranes **132** made of fibers or microfibers (for example, glass fibers) **134** as illustrated in FIG. **2**. The fibers or microfibers **134** (called herein "the fibers") may be randomly distributed within transport layer **132** and some of these fibers are interconnected to each other so that the empty spaces **136** between them form mass transport paths or microchannels **138**. FIG. **1**B shows two microchannels **138** as an example. A large number of microchannels is formed in each transport layer **132**. In one application, transport layers **132** may be made of any porous material that has plural pores sized so that the saline feed is pushed by capillarity upwards, toward the solar absorber layer **120**. For the capillarity to be present, an average diameter of the microchannel, tube, pore or empty spaces **136** should be 6 mm or less. Thus, any material that has microchannels, tubes, pores, or empty spaces **136** with an average diameter smaller than 6 mm can be used as the material for the transport layer **132**. The transport layer **132** may have any thickness and/or width. However, a length of the transport layer **132** is limited as discussed later.

[0035] FIG. **1**B illustrates the salt backflow **152** through a single mass transport path **138**. Note that the path **138** in FIG. **1**B is schematically illustrated to be a straight channel. However, in practice, the path **138** is expected to have a more wavy shape, as the path **138** is formed by the empty spaces **136** shown in FIG. **2**, and those spaces are not typically aligned along a straight line. However, if the material used for the transport layer **132** is, for example, a type of clay with plural pores, than those pores (paths **138**) may be engineered to follow a straight line. The shape of the path **138** can vary in many ways as long as the path **138** extends from the saline feed **140** to the solar absorber **120** and supports capillarity.

[0036] FIG. **1**B shows that salt molecules **150** travel from the saline feed **140** toward the solar absorber layer **120**. As more salt molecules **150** accumulate near the solar absorber layer **120**, their concentration at the top of the path **138** increases. The increased concentration of the salt molecules **150** at the top of the path **138** make the salt molecules to travel back toward the saline feed **140**, as schematically illustrated by arrow **152** in FIG. **1**B. This salt backflow **152** prevents the salt accumulation at the solar absorber layer **120**, which is one of the biggest problems of the existing evaporator systems. Because of the presence of the salt backflow **152**, the system **100** can function for a long period of time without the need of being cleaned and without the risk of losing its efficiency.

[0037] At the same time, part of the heat accumulated at the solar absorber layer **120** diffuses along path **138**, toward the saline feed **140**. Due to the choice of the vertical length L of the path **138**, the heat from the solar absorber layer **120** is essentially entirely transformed into vapor **154** along the path, so that practically no heat reaches the saline feed **140**. This is advantageous because the bulk water **140** does not need to be heated and the vapor **154** generated along the path **138** further evaporates the water from the saline feed that circulates through the path. Thus, the arrangement shown in FIG. **1**A simultaneously achieves salt rejection from the solar absorber layer **120** and vapor generation through direct heat and heat recycling.

[0038] The main source of heat for vapor generation is the light **160** from the sun and this light is transformed into heat by the solar absorber layer **120**. For this reason, the solar absorber layer **120** is placed on top of the system **100**, to be exposed to the sun as much as possible. The solar absorber layer **120** may be made of various materials, for example, carbon nanotubes (CNT), or partially oxidized carbon nanotubes, or other known materials that transform light into heat. The CNT **122**, which are schematically illustrated in FIG. **3**, are formed onto the solar absorber layer **120** and this layer is placed on top of the plural paths **138**, as schematically indicated in FIG. **1**B. Thus, in one embodiment, the solar absorber layer **120** is in direct contact with the top portions of the paths **138**.

[0039] The support frame **110** includes a top body **112** and one or more support pillars **114**, as shown in FIGS. **4**A to **4**C. Although these figures show the top body **112** being supported by four pillars **114**, one skilled in the art would understand that only one central pillar may be used, or three or more than four pillars as long as the pillar supports the top body **112** above the saline feed **140**. FIG. **4**A, which is a top view of system **100**, shows the entire solar absorber layer **120** extending over the top body **112**, so that no transport layer **132** is visible. FIG. **4**B, which is a front view of system **100**, shows plural transport layers **132** being supported by the top body **112** and a bottom body **116** so that the plural transport layers **132** are parallel to each other and vertical. Note that each of the top and bottom bodies have corresponding slots that receive transport layers **132**. Layers **132** may be simply fitted into the corresponding slots or may be glued to the slots. The number of transport layers **132** can vary, as discussed later. Macrochannels **139** are formed by two adjacent transport layers **132**. A width W of such macro channels may be selected to be less than 6 mm, so that capillarity may also appear in the macrochannels **139**, not only within the microchannels **138**. This is advantageous as the vapors **152** escaping the microchannels **138** directly interact with the saline particles in the macro channels and further generate vapors. At this step the heat lost from the solar absorber layer is recycled as this heat is not lost to the bulk water, but rather it is used to further generate vapors. FIG. **4**C shows a side view of system **100**, with only one single transport layer **132** being visible. In this embodiment, the solar absorber layer **120** is a square, having a side Ls of about 30 mm, and a side LL of the frame **110** is about 40 mm, with a lateral size S of each pillar **114** being about 5 mm. A height H of the frame **110** is about 35 mm. Note that the term "about" is used in this application to indicate a variation of up to 20% of the reference number that is characterized by the term "about." A thickness t of the transport layer **132** may be about 0.45 mm. FIG. **4**C also shows the addition of a cover **410** around the top part of the transport layers **132**, where the vapors **154** are mainly generated. The cover **410** is shown in this embodiment being attached to the pillars **114**. However, in a different embodiment, the cover may be attached to the top body **112** or any other part of the support frame **110** or the mass and heat transport component **130**. The cover **410** is designed to promote the condensation of the water vapor **154** and generate condensate **156**, as schematically illustrated in the figure. The cover **410** may be made of any material that is transparent to the light so that the light can reach the solar absorber layer **120**.

[0040] FIG. **4**B shows an effective length L of one or more of the transport layers **132** being considered to be between the top and bottom bodies **112** and **116** of the support frame **110**. In other words, although the actual length l of the transport layers is longer, l>L, as they extend past the bottom body **116**, so that when the system **100** is placed above or to float in the saline feed **140**, the bottom parts of the transport layers **132** are fully immersed in the feed, the effective length L of the transport layers, for calculation purposes, is considered herein to not be the actual length l, but an effective length.

[0041] FIG. **5** illustrates a salt-rejection evaporator system **500** that includes plural systems **100-1**, where I is an integer equal to or larger than 1, and a common support frame **110** holds all the transport layers **132** of each system **100-1**. The bottom body **116** is placed onto a floating component **502**, for example, made of a plastic or other material that is lighter than the saline feed **140**. In this case, the saline feed **140** may be sea or ocean water and the system **500** may be floating on it. In this embodiment, the top part of transport layers **132** extends past the top body **112**, as shown in the figure. Note that in FIGS. **1**A, **4**B, and **4**C, the top of the transport layers **132** is flush with the top body **112**. Further, FIG. **5** shows a dome (or cover) **510**, which is made of a light material, which covers the entire top portions of the transport layers **132**. Dome **510** is made of a material that is transparent to light, so that the solar light reaches the solar absorber layer **120** almost unimpeded. The vapors **154**, which are generated in and between the paths **138** of the transport layers **132**, raise and condense on the walls of the dome **510**, forming the condensate **156**, which is collected at the bottom of the dome. An output port **512** may be formed in a wall of the dome **510** to collect the condensate **156**, which is essentially desalinated water. While FIG. **5** shows the system **500** including plural systems **100**, it is possible to have only one system **100**.

[0042] The embodiments discussed above show transport layers **132** being parallel to each other and essentially being arranged to be vertical, i.e., perpendicular to the top or bottom bodies **112** and **116**. However, in another embodiment, as illustrated in FIG. **6**A, the transport layers **132** are still parallel to each other, but inclined with a non-zero angle α relative to the vertical g. Note that the system **100** is shown herein with the dome **500**, thus forming a chamber **502** around the top portions of the transport layers **132**. In yet another embodiment, as illustrated in FIG. **6**B, transport layers **132** are not flat and extending in a single plane, they are wavy as they extend from the bottom body **116** to the top body **112**. Layers **132** may have any shape in this embodiment. It is noted that the height H of the transport layers **132** between the top and bottom bodies **112**, **116** remains the same as in the previous embodiments, although the effective length L and the actual length l of these layers is larger than in the previous embodiments.

[0043] A method for manufacturing the salt-rejection evaporator system **100** is now discussed with regard to FIGS. **7**A to **7**D. The method starts in FIG. **7**A with the manufacturing of the top body **112**, pillars **114**, and the bottom body **116**. In one application, these parts of the support frame **110** are 3D printed. In yet another embodiment, they are made of a polymethyl methacrylate (PMMA) plate via laser cutting. The parts are then bonded in step **702** to obtain the support frame **110** shown in FIG. **7**B. Note that each of the top and bottom bodies **112** and **116** have corresponding slots **118** for receiving the transport layers **132**. In step **704**, transport layers **132** are inserted into the corresponding slots formed in the top and bottom bodies, to form the mass and heat transport component **130**. A transport layer **132** may have an effective length L of about 3 cm and a thickness t of about 0.45 mm. In step **706**, the solar absorber layer **120** is placed over the top body **112**, to be in direct contact with the top of the transport layers **132**. If the transport layers **132** are flush with the top body **112**, then the solar absorber layer **120** is in direct contact with both the top parts of the transport layers **132** and the top body **112**. However, if the transport layers **132** extend past the top body **112**, then the solar absorber layer **120** is in direct contact only with the transport layers **132**. The solar absorber layer **120** was fabricated in this embodiment by loading partially oxidized CNTs on transport layers **132**. The reason for choosing partially oxidized CNTs over pristine CNTs is that their hydrophilicity facilitates the formation of a uniform coating on the transport layers **132**. For example, about 3 g of CNTs (diameter: 110-170 nm and length: 5-9 μm) was dispersed in a 120 ml acid mixture (90 ml H_2SO_4+30 ml HNO_3) and reacted at 70° C. for 5 h. The resultant product was collected by filtration and washed until neutralization. Next, a certain amount of partially oxidized CNTs was dispersed in water through ultrasonication and then filtered through a glass fiber membrane. The obtained solar absorber was dried at 60° C. The CNT loading percentage was determined to be ~11 wt. %.

[0044] For the conventional salt-rejection solar evaporation systems, water evaporation is confined to the solar absorber surface, and the salt backflow is accompanied by an undesired heat dissipation from the solar absorber to the bulk water (saline feed), thus resulting in a low evaporation rate. This limitation of the existing devices is solved by the 3D evaporator systems **100/500** because transport layers **132** connect the saline water to the solar absorber layer, they have hydrophilic microchannels that can pump the saline water to the solar absorber layer via a capillary force, and they also promote the flow back of the excessive salt into the saline feed through the brine-filled microchannels (i.e., paths **138**) via diffusion and convection, as schematically illustrated in FIG. **1**B. The adequate mass transfer via a high density of transport layers **132** ensures a continuous water supply and an efficient salt backflow, thus enabling a unique salt rejection capability. Unlike the conventional salt-rejection systems, where the heat conducted from the solar absorber to the bulk water is simply dissipated and considered "wasted," the transport layers **132** can efficiently recover this conductive heat to generate additional vapor from the brine flowing through their microchannels **138**. The microchannels **138** within the layers **132** and the macro-channels **139** between the spaced layers **132** together form a highly open structure that allows the generated vapor to be easily released from the layers **132**'s surfaces, in all directions. Thus, by selecting the height of the layers **132** to have a value in a specific range (which is discussed later), the conductive heat can be largely confined within the mass and heat transport component **130** for vapor generation, thereby significantly improving the water evaporation efficiency of the system.

[0045] The solar absorber layer **120** of the system **100/500** was found to have a solar absorption of about 96% as shown in FIG. **8** because of the porous fibrous light-trapping structure (see FIG. **3**) and the inherent black property of the CNT. Considering their abundant hydrophilic microchannels formed by intertwined glass fibers (see FIG. **2**), the glass fibers **136** were selected for making up the layers **132**. A glass fiber made transport layer **132** can immediately absorb a water droplet upon touching it because of its high affinity to water. Moreover, vertically aligned layers **132** can pump water to a 25 cm height in 60 min, demonstrating its strong capillary force for water transfer.

[0046] The advantages of the system **100/500** discussed above over the existing devices are achieved due to the configuration/structure of the system, but also, in part, due to the effective length L (not the actual length l) of layers **132**. The effective length L was determined, in one embodiment, as now discussed. To avoid salt crystallization at or near the solar absorber layer **120**, excess salt must be efficiently transported back toward the saline feed **140** to maintain the top surface's salinity below the saturation point. In this system, salt can be rejected via diffusion and convection through the brine-filled microchannels **138** under the driving force of the concentration gradient (osmosis) and gravity. The mass flow rate (J) through the system **100/500** can be described by the diffusion-convection equation as follows:

$$J = J_{diff} + J_{conv} = nA\varepsilon\left(\frac{k_d(C_{evp} - C_0)}{H} + k_c(\rho_{evp} - \rho_0)\right) \tag{1}$$

[0047] where J_{diff} and J_{conv} are the mass flow rate caused by diffusion and convection, respectively; n is the number of layers **132**; A, ε, and Hare the cross-section area, porosity, and height of the layers **132**, respectively; K_d and k_c are the diffusion and average convective coefficients of salt, respectively; C_{evp} and C_0 are the salt concentrations on the evaporation surface and in the bulk saline water, respectively; and ρ_{evp} and ρ_0 are the salt solution densities on the evaporation surface and in the bulk saline water, respectively.

[0048] In Equation (1), the mass transport rate is proportional to the number n of transport layers **132**. This relation was validated by fabricating systems **100** with varying numbers of transport layers **132**, ranging from 2 to 32, as illustrated in FIG. **9**. For this experiment, the cross-section area (A) of the transport layer **132** is about 0.135 cm^2, its height (H) is about 3 cm and its porosity (E) is about 65%. The evaporation for all these samples was evaluated using high-salinity water (10 wt. % NaCl). The evaluation was performed under 1 sun illumination for 12 h. The inventors found that salt crystals massively accumulated on the 2-bridge evaporator surface because of its insufficient mass transfer. This salt accumulation was mitigated with the increase in the transport layers **132** number. For the evaporator system **100** containing 32 layers **132**, no salt crystals were observed on the surface, after the 12 h operation. At an insufficient number of transport layers **132** (e.g., 16 layers), the evaporation rate gradually decreased as the vapor generation progressed because of the increased evaporation surface salinity. In this regard, FIG. **10** shows the corresponding mass change curves for the system samples **100** shown in FIG. **9**. It is noted that when the number of transport layers **132** reaches a value of about 32, the excess salt can be efficiently rejected to maintain the evaporation surface at a relatively low salinity. Remarkably, the evaporation rate of the 32-transport layers evaporator system was found to be about 1.44 $kg/m^2/h$ without degradation during the 12 h operation.

[0049] Subsequently, the inventors performed a complementary experiment to further demonstrate the salt backflow introduced by the 32-transport layer evaporator system **100**. In this experiment, the evaporator system was placed in a high-concentration saline water (10 wt. % NaCl solution) and exposed to 1 sun illumination, and 1 g of NaCl salt was added on its surface (not shown). It was observed that during vapor generation, the added salt was gradually dissolved and completely removed in about 11 h. This experiment demonstrated that the salt backflow rate of the 32-transport layer evaporator system in the 10 wt. % NaCl solution was higher than the salt generation rate, thus confirming the salt rejection feature of the proposed architecture. When the saline feed's salinity was further increased, to test the maximum applicable salt concentration of this evaporator system, the inventors observed that because of the effects of diffusion and convection, the salt backflow decreased as the salinity (i.e., C_0 and ρ_0) increased, and salt started to crystallize at the edges of the solar absorber after 12 h operation when 14 wt. % NaCl solution was used for the test. Based on the corresponding evaporation rate, the salt backflow along the transport layers **132** was calculated to be about 1.1 $g/cm^2/h$. Interestingly, this unique mass transport feature is intertwined with its heat transport feature, as now discussed.

[0050] For the heat transport investigation, the inventors fabricated 32-transport layers evaporator systems with different heights, as illustrated in FIG. **11**, and evaluated their evaporation performance. Under dark conditions, the evaporator system without transport layers **132** (i.e., transport layer height: 0 cm) exhibited a natural evaporation rate of 0.15 $kg/m^2/h$. The natural evaporation rate increased with the incorporation of the transportation layers due to the increased surface area. Specifically, it linearly increased by about 0.04 $kg/m^2/h$ for every 1 cm increase in the height of the transport layers **132**. Under 1 sun illumination, the evaporation rate of the evaporator system without transport layers was only 0.99 $kg/m^2/h$ because of the massive conductive heat dissipation to the bulk water, as shown in FIG. **12**. The transport layers usage considerably promoted solar evaporation. The evaporation rate increased to 1.58-1.73 $kg/m^2/h$ when the transport layer height reached 2-5 cm, as also shown in FIG. **12**. These values are even higher than the theoretical upper limit for solar evaporation (~1.44 $kg/m^2/h$), which can be attributed to the natural evaporation contribution. When the transport layers' height exceeded 3 cm, the evaporation rate increased by about 0.04 $kg/m^2/h$ for every 1 cm increase in height (see FIG. **12**), which was consistent with the result obtained under dark conditions. This consistency suggests that the 3 cm height is sufficient for the transport layers **132** to maximize solar evaporation (note that additional increase in the transport layers height only increases natural evaporation).

[0051] The energy loss channels for the evaporation system **100** primarily include conductive heat loss into the bulk water ($P_{cond.}$), radiative heat loss ($P_{rad.}$), and convective heat loss to the environment ($P_{convec.}$). Therefore, the power flux available for evaporation (P_{evp}) can be described as follows:

$$P_{evp} = P_{solar} - P_{cond.} - P_{rad.} - P_{convec.}, \quad (2)$$

[0052] where the solar energy input $P_{solar}=\alpha C_{opt} q_i$; α is the light absorption coefficient; C_{opt} is the optical concentration; and q_i is the direct solar illumination. The conductive heat flux $P_{cond.}=k(T_{sa}-T_{bw})/l$, where k is the thermal conductivity; T_{sa} and T_{bw} are the temperatures of the solar absorber and the bulk water, respectively; and l is the heat conduction path of the transport layer **132**, which may be equal to the effective length L of this layer or even longer for the embodiments shown in FIGS. **6**A and **6**B. The radiative heat flux $P_{rad.}=\varepsilon\sigma(T_1^{\,4}-T_2^{\,4})$, while the convective heat flux $P_{convec.}=h(T_1-T_2)$, ε is the optical emission, ε**0** is the Stefan-Boltzmann constant, h is the convection heat transfer coefficient, and T_1 and T_2 are the temperatures of the evaporator and environment, respectively.

[0053] The energy loss caused by the heat transfer from the top surface of the solar absorber layer **120** to the saline feed **140** (i.e., $P_{cond.}$) can be minimized by increasing the height of the transport layer (i.e., l) to confine the conductive heat within the transport layers. This effect was visualized using infrared imaging to display the temperature gradients along the transport layers with different heights. The results showed that the temperature at the bottom of the evaporator was similar to the ambient temperature when the transport layers height reached or exceeded 3 cm.

[0054] Next, the inventors recorded the internal temperature variation at different distances from the solar absorber layer **120** under solar illumination. The results showed that the temperature stabilized after 60 min, when the internal temperature at 3 cm to the solar absorber layer was similar to that of the surrounding environment (see FIG. **13**), indicating that the conductive heat was completely confined in the top 3 cm of the transport layers **132**. This confinement effect was also demonstrated by the temperature change of the bulk water. For the evaporator system without transport layers, the bulk water temperature increased from ~21 to ~26.2° C. after a 3 h operation due to the continuous heat input; for the evaporator system with 3-cm transport layers, however, the bulk water temperature was maintained at room temperature (~21.3° C.), thus confirming the suppression of heat dissipation into the saline feed. Thus, the effective length L of the transport layers is selected to suppress heat loss to the saline feed. While this length L is about 3 cm in this embodiment, one skilled in the art would understand that this length can be smaller or larger depending on the type of fibers used in the transport layer, the size of the microchannels, the composition of the saline feed, the ambient temperature, etc.

[0055] Novel in the system **100**, the confined heat energy is exploited to generate additional vapor from the transport layers **132** surfaces, as schematically illustrated in FIG. **1**B, which can be efficiently released via the highly open inter-bridge spaces. To reveal this additional vapor generation from the external surfaces of the transport layers **132**, an evaporator system **100** having 32 transport layers (3 cm high) was used to perform a control experiment. In this experiment, the evaporator body was enclosed with an airtight polypropylene film, thus leaving only the upper surface exposed to the open space for vapor release. After a 3 h operation, many water droplets condensed on the inner film surface, thus confirming that the transport layers released vapor. Compared to the completely open evaporator system, the evaporation rate of the partially enclosed system decreased by about 31%, demonstrating the importance of the open-channel design for enhanced interfacial evaporation.

[0056] Furthermore, the inventors performed a cycling experiment to evaluate the evaporator system's stability. In each cycle, the evaporator system ran for 12 h under 1 sun illumination and in a dark environment for another 12 h to simulate day and night alternation. FIG. **14** shows that during this long-term test (with 10 wt. % NaCl solution), the mass change of the NaCl solution in each cycle linearly evolved and the evaporation rate stabilized at about 1.44 $kg/m^2/h$. No performance degradation was observed after a seven-day cycling experiment.

[0057] Compared to the previously reported salt-rejection evaporators (evaporation rate: from 1.24 to 1.28 $kg/m^2/h$ for 10 wt. % NaCl solution) [5, 6, 7], the evaporator system **100** demonstrated a higher evaporation rate under similar conditions due to the heat confinement effect and the natural evaporation contribution. However, high evaporation efficiency alone is not sufficient for water production applications. If the evaporated moisture is not collected, it can only be considered as a pollutant to the environment considering that it has the greatest greenhouse effect among various components in the atmosphere. Water collection, which is equally important as vapor generation, has been largely ignored in many previous studies on salt-rejection evaporators.

[0058] Therefore, part of the evaporator system **100** shown in FIG. **1**A was enclosed with the transparent cover **410** as shown in FIG. **4**C or dome **510** as shown in FIG. **5**. The cover and/or dome were made of polymethyl methacrylate (PMMA) plates, and they made the system **100/500** to produce water by condensing the evaporated moisture. The inventors investigated the effects of transport layer number and height on the water production capacity of this system. When the transport layer height was fixed at 3 cm, the amount of collected water increased with the number of transport layers, which is consistent with the observation in the open system, confirming that the enhanced salt backflow facilitates water evaporation. When the transport layer number was fixed at **32**, the amount of collected water increased with the transport layer's effective height and reached the maximum at 3 cm, while further increasing the height did not produce more water. These results are consistent with the conclusion above that a 3 cm height is sufficient to confine the conductive heat while further increasing the height only increases natural evaporation that does not contribute to water production. According to three-hour test results, the water production rate of the enclosed evaporator system in the optimal configuration (32 transport layers; 3 cm high) was calculated to be about 0.68 $kg/m^2/h$.

[0059] The inventors also investigated the water generation performance of the enclosed system under different salinity conditions using NaCl solutions (3.5 wt. % to 20 wt. %). The results showed that the water production efficiency monotonically decreased from about 0.73 $kg/m^2/h$ for 3.5 wt. % NaCl solution to about 0.63 $kg/m^2/h$ for 20 wt. % NaCl solution. The relatively low water production efficiency associated with the high-salinity brines is mainly due to their low saturated vapor pressure, and partly due to the decreased photothermic conversion efficiency caused by salt precipitation. For instance, when using brine containing 20 wt. % NaCl, salt precipitation emerged at the periphery of the evaporator system after three hours of testing.

[0060] A fabricated solar-driven water generation system **100**, having a 15×26 cm evaporator area was tested on the roof of a building and also on the sea surface. In the first test, discharged water from an RO system was used as the source water (salinity: ~8.7%). The daily evaluation started at 8:00 and ended at 17:00. As shown in FIG. **15**, the evaporator surface was heated by solar light to a temperature 4-15° C. higher than the environment. However, the temperature at the transport layer's bottom was almost the same as the environment temperature, indicating that the conductive heat was confined, with only a small amount transferred to the saline feed. Consequently, saline water can be efficiently evaporated and condensed at the cover surface for the water collection. The total collected water was about 175 ml. Based on the evaporator area (390 cm^2), the daily water productivity was calculated to be about 5.0 L/m^2. The inventors measured the ion contents of the water samples to evaluate the water quality. Compared with the discharged water from the RO plant, the ion concentration of the condensed water was reduced by at least four orders of magnitude, thus fully meeting the WHO drinking water requirements.

[0061] A continuous test was performed over five days to evaluate the system's performance stability. The daily water collection rate fluctuated in the range of 4.7-5.2 L/m^2, depending on the specific solar insolation of the day. The corresponding solar-water collection efficiency was 39%-42%. Remarkably, no salt accumulation was observed during this five-day outdoor operation. These results demonstrate the potential of the fabricated evaporator system to extract freshwater from the wastewater discharged by RO plants.

[0062] Next, the same evaporation system was tested in a floating configuration in the Red Sea (salt content: ~4.3%) to demonstrate its potential for practical seawater desalination. The test lasted for five days. It was found that the daily freshwater productivity ranged from 5.0 to 5.8 L/m^2 with a stable solar-water collection efficiency of 42%-45%, which was consistent with the rooftop test. This freshwater productivity was approximately two times higher than the previous record of the salt-rejection solar evaporator discussed in [5] (~2.5 L/m^2 per day). The field test demonstrated a high-performance solar evaporator system that could help in disaster relief or strengthen the resilience of individuals living on boats, coastal areas, or next to saline sources of water.

[0063] While the systems **100/500** discussed above use a saline source of water, it is possible, based on a similar architecture, to make a system that uses atmospheric water for producing the condensate. The solar-driven atmospheric water extraction (AWE) is a sustainable technology with great potential for decentralized freshwater supply. However, most AWE systems can only produce water intermittently, under sunlight, due to the cyclic nature of sorption and desorption, and their widespread adoption is hindered by complex design requirements or periodic manual operations. In the following embodiments, a fully passive AWE system is introduced that continuously produces freshwater under sunlight, without the need for additional maintenance or power. By adapting the three-dimensional architecture of system **100**, with abundant microfibrous structure to facilitate spontaneous mass transport and efficient photothermal energy utilization, this system can consistently produce, in one embodiment, 0.65 $L/m^2/h$ freshwater under 1 sun illumination at 90% relative humidity (RH) and can even function in arid environments with RH as low as 40%. The practical performance of a scaled-up system was tested over a total period of 35 days across two seasons, achieving freshwater production of 2.0-2.9 L/m^2/day during a 10-day summer test and 1.0-2.8 L/m^2/day during a 25-day fall test, without requiring additional maintenance. These embodiments demonstrate the potential of the system for off-grid, point-of-use irrigation applications by growing plants with atmospherically collected water. This passive AWE system, which uses solar energy to continuously extract moisture from the air for drinking and irrigation, is a promising solution to address the intertwined energy, water and food challenges, especially for remote and water-scarce regions. Note that system **100** uses no external energy, i.e., only the light received from the sun, no batteries, no piezoelectric device, no fuel cell, etc. Thus, the system **100** not only requires no man-made energy, but also has a zero CO_2 footprint as the sun light is directly and totally transformed into heat.

[0064] Atmospheric water is an omnipresent natural resource that is estimated to be six times the total freshwater volume in rivers worldwide. Furthermore, its availability is expected to increase due to global warming. Sorption-based AWE using sunlight provides a sustainable strategy for decentralized freshwater supply, which is particularly important for remote, water-stressed regions. Typically, hygroscopic sorbents are employed to extract moisture from the surrounding environment [8, 9]. Once these sorbents become saturated, the system is sealed and exposed to sunlight to initiate the release of the captured water, thus enabling the production of freshwater. Due to the slow kinetics of water capture and release by the sorbent/sorbent bed, many previous systems are limited to only one sorption-desorption cycle per day, with moisture capture occurring at night and water production taking place during the daytime. As a result, the productivity of these systems is inherently constrained by the adsorption capacity of the sorbent. For instance, the reported water productivity for a pioneering system is around 0.77 L/m^2/day. Other sorption-based prototypes have produced as little as 0.1 L/kg/day (i.e., one kg of water sorption materials like metal-organic framework can produce only 100 mg water per day).

[0065] To overcome these limitations, multiple-cycle systems have been proposed through the development of sorbent/sorbent beds with rapid kinetics. Despite promising developments, the widespread adoption of this technology is still constrained by the high cost of nanomaterials and the challenges associated with scaling up prototypes. Moreover, due to the cyclic nature of the systems, they can only produce water intermittently, under sunlight, and the switching of cycles necessitates the use of active system or labor-intensive operation and auxiliary moving parts, resulting in energy-intensive process and system complexity. To fully exploit the vast potential of AWE, a truly passive and scalable system capable of efficiently and continuously producing freshwater is needed.

[0066] FIG. **16** shows a solar-driven atmospheric water extraction system **1600** that has a support frame **1610**, the solar absorber layer **120**, the mass and heat transport component **130** having the plural transport layers **132**, a sorption system **1620**, and a cover **1630**. The sorption system **1620** includes a tank **1622** that is configured to hold a sorption liquid **1624**. The plural transport layers **132** extend into the sorption liquid **1624** and based on capillarity, push the sorption liquid, along the microchannels **138**, toward the solar absorber layer **120**. Cover **1630** is located around the top of the plural transport layers **132**, similar to the system **500** illustrated in FIG. **5**, for forming chamber **1632**, for promoting the condensation of the vapors and the collection of the condensate. In one application, cover **1630** is identical to the cover **410** or dome **510**.

[0067] The microchannels **138** (not shown in this figure, but identical to those shown in FIG. **1**B of the transport layers **132**) are infused with the liquid sorbent **1624**. Depending on the temperature distribution, which is schematically illustrated in FIG. **17** by arrow **1701**, the transport layers **132** are divided into two functional regions: the room-temperature region **1702**, which is exposed to the environment for continuous atmospheric water capture, and the high-temperature region **1704**, which is enclosed in the chamber **1632** for freshwater generation. During operation, the room-temperature region **1702** captures atmospheric water and stores the captured water **1640** in the tank **1622**, if no sunlight is available. When system **1600** receives sunlight, the solar absorber layer **120** converts the light to heat and generates concentrated vapor **154** in the high-temperature region **1704**. Vapor **154** released in the process condenses on the cover **1630'** wall, producing freshwater **156**. The captured atmospheric water **1640**, stored in tank **1622**, is transported to the high-temperature region **1704** via capillarity, as schematically illustrated by arrow **1708**, ensuring uninterrupted and efficient vapor generation. Simultaneously, the concentrated liquid sorbent **1624** in the high-temperature region **1704** is transported back to the room-temperature region **1702** through diffusion and convection **1710**, where it can continue to capture atmospheric water, as schematically illustrated in FIG. **17**. This process results in completely passive and maintenance-free atmospheric water production. By directing the condensed droplets **156** to plant roots (not shown) located next to the system **1600**, this system can facilitate off-grid irrigation using only atmospheric water.

[0068] In one implementation, the system **1600** has solar absorber layer **120** made by loading partially oxidized carbon nanotubes (CNTs) onto glass fiber membranes (GFM) that form the transport layers **132**. Thanks to the light-trapping microstructures and the inherent black property of CNTs, the solar absorptance of the obtained solar absorber can reach ~96% in the wet state. The GFM was also selected for the fabrication of the transport layers due to its intertwined fibrous structure. At the microscale, the intertwined fibers create abundant capillary microchannels, giving the GFM a strong water transport capability. A particularly noteworthy observation is the linear increase in water uptake with the height of capillary rise in the microchannels **138**. This finding implies that nearly all the microchannels are saturated with water, and the interconnected, water-filled channels provide a path for the backflow of sorbents.

[0069] In this embodiment, the inventors selected a lithium chloride (LiCl) solution as the hygroscopic liquid sorbent **1624** due to its availability, cost-effectiveness, wide range of applicable relative humidity (RH), and strong water molecule capture capability. The saturated LiCl solution can capture water molecules at RH as low as 15%, and the adsorption capacity reaches ~2.5 g/g when the RH increases to 90%. Moreover, the water trapped by the LiCl solution can be efficiently released without hysteresis as the RH decreases. Different from the support frame **110** of the system **100**, the support frame **1610** has the tank **1622** attached to the bottom body **116**, and the poles **114** may be either attached to the body **116**, or directly to the tank **1622**. A method for making system **1600** is similar to the method for making the system **100**, which was discussed in FIGS. **7**A to **7**D. Thus, the method is not repeated herein.

[0070] The AWE system **1600**'s performance was tested, similar to the system **100**. First, system **1600** was tested within a controlled environment. When the system was exposed to solar radiation, the solar absorber layer **120** converts the incoming light into heat energy. This energy is then conducted along the transport layers, creating the temperature gradient **1701**. To achieve efficient and stable water production, the number of transport layers has been selected to be 32 and the height of the vapor generation zone **1702** was selected to be about 3 cm. Similarly, to facilitate fast atmospheric water capture kinetics, the height of the atmospheric water capture zone **1704** was selected to be about 5 cm. Thus, the effective height L of the transport layers **132** is about 8 cm in this embodiment, with 3 cm of these layers being located within the cover **1630** and 5 cm of the layers extending between the cover and the bottom body **116**. The sorbent solution was selected to be 0.24 g/g LiCl solution at 90% RH. In the absence of sunlight, the system **1600** engaged in atmospheric water capture, with the collected water stored in the tank **1622**. As the process continued, the water level in the tank **1622** gradually rose. Upon exposure to sunlight, vapor was released and subsequently condensed to produce condensate water. Simultaneously, a portion of the stored water was transported upward to compensate for the reduced water content in the vapor generation zone **1704**, resulting in a corresponding decrease in the water level within the tank **1622**. Interestingly, the decrease in the water level was not significant, as the system **1600** continued to capture atmospheric water during the water production process. To highlight this unique feature of the system **1600**, i.e., simultaneous water capture and production, the inventors monitored the weight changes of the system and the generated water amount during its operation. After running for 8 h under 1 sun illumination, the system produced about 1.8 ml of water while capturing about 1.5 g of atmospheric water from an environment with a RH of about 65%. Throughout the operation, the water production rate stabilized around 0.22 $L/m^2/h$.

[0071] To further assess the AWE potential of the system **1600**, the inventors conducted an 8-day consecutive water production evaluation in a laboratory setting under various RH conditions, operating in a maintenance-free mode. In this operating mode, the system operated independently, without requiring any maintenance or adjustment to switch the water capture and water production modes. Each day, the system underwent 16 hours of water capture followed by 8 hours of water production under 1 sun illumination. Throughout the process, the water molecules and sorbents within the system gradually reached a mass transport equilibrium, resulting in stable water production. As RH increased from 60% to 90%, the stabilized water production rate under 1 sun illumination rose from about 0.04 $kg/m^2/h$

at 60% RH to about 0.65 kg/m^2 at 90% RH. This demonstrates the potential of the system to extract fresh water from relatively humid air.

[0072] The inventors also assessed the water production of the proposed system in a manual mode, in which the operator manually opened the chamber **1632** during the water atmospheric water capture process to accelerate mass transport. In this operating mode, the system can operate at RH as low as 40%, with stabilized water production rate reaching 0.68 $L/m^2/h$ at 90% RH. Consequently, the system can operate at lower RH and is applicable in most parts of the world using this conventional operating mode.

[0073] To test the outdoor performance of the AWE system **1600**, it was placed on a rooftop. The system operated with water capture occurring during the nighttime. After 24 hours of operation, the weight of the water produced was measured. It generated approximately 95 ml of water. Based on the projected area of the system, the water production was calculated to be about 2.9 $L/m^2/day$.

[0074] The inventors further evaluated the water production capability of the system over 35 days across two seasons in Thuwal, Saudi Arabia: 10 days in summer and 25 days in fall. During the summer, with strong sunlight and high temperatures, daily water production ranged between 65-96 ml (equivalent to 2.0-3.0 $L/m^2/day$). This was influenced by both the received solar energy and RH conditions. In the fall, solar intensity and temperature both decreased, yet the prototype remained functional, with daily water production varying between 35 and 90 ml (equivalent to 1.1-2.8 $L/m^2/day$). Due to its completely passive working principle, this system operates spontaneously and continuously to produce water without requiring further maintenance.

[0075] Because the AWE system **1600** produces water by extracting it from air and is exposed to environment throughout the entire operation, the collected water may easily be contaminated by airborne pollutants, which were often observed in other atmospheric water harvesting technologies such as fog collection and dew condensation. Therefore, the quality of the water produced by the system was measured. The inventors analyzed the concentrations of ions and microbial cells in the produced water. All ionic indicators were found to be well below the guideline values of WHO. For microbiology, all values of HPC 36° C., total coliforms and *Escherichia coli* were below the detection limits. The measure for active biomass was below the detection limit of the method, indicating very low or no bacterial presence. Thus, the water quality generated by the system **1600** meets the requirements for drinking and irrigation use.

[0076] The AWE system **1600**, which requires no bulk water source and minimal installation and maintenance, enables autonomous, point-of-use irrigation. This represents a revolutionary solution for irrigation in semi-arid and arid regions lacking liquid water resources. The disclosed embodiments provide a water generation system that uses plural transport layers for moving water from a source to a solar absorber layer. The transport layers are configured, as discussed in the above embodiments, to ensure salt/sorbent backflow so that the device is maintenance free, and also to reuse the convection heat without heating the water source. It should be understood that this description is not intended to limit the invention. On the contrary, the embodiments are intended to cover alternatives, modifications and equivalents, which are included in the spirit and scope of the invention as defined by the appended claims. Further, in the detailed description of the embodiments, numerous specific details are set forth in order to provide a comprehensive understanding of the claimed invention. However, one skilled in the art would understand that various embodiments may be practiced without such specific details.

[0077] Although the features and elements of the present embodiments are described in the embodiments in particular combinations, each feature or element can be used alone without the other features and elements of the embodiments or in various combinations with or without other features and elements disclosed herein.

[0078] This written description uses examples of the subject matter disclosed to enable any person skilled in the art to practice the same, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the subject matter is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims.

REFERENCES

[0079] The entire content of all the publications listed herein is incorporated by reference in this patent application.

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1. A salt-rejection evaporator system comprising:

a support frame;

a mass and heat transport component supported by the support frame, the mass and heat transport component having plural transport layers; and

a solar absorber layer located on top of the transport layers, wherein the plural transport layers include plural microchannels that support capillarity, promote a flow of a saline feed toward the solar absorber layer and generate vapors due to heat generated by the solar absorber layer, and

wherein the solar absorber layer is formed directly on top of the plural transport layers.

2. The system of claim **1**, wherein the plural transport layers are parallel to each other.

3. The system of claim **2**, wherein the plural transport layers extend along a direction that is perpendicular to the solar absorber layer.

4. The system of claim **1**, wherein the plural transport layers include 32 layers and an effective length of the plural transport layers is about 3 cm.

5. The system of claim **1**, wherein the plural transport layers are made of glass fibers and the solar absorber layer is made of carbon nanotubes.

6. The system of claim **1**, wherein an effective length L of the plural transport layers is selected so that a temperature increase generated at the solar absorber layer, due to the transformation of light into heat, is not heating a saline feed at a bottom of the plural transport layers.

7. The system of claim **1**, wherein heat generated by the solar absorber layer heats a saline feed that flows through microchannels of the plural transport layers and generates vapors.

8. The system of claim **7**, further comprising:

a cover located over a top portion of the plural transport layers, the cover being configured to condensate the vapors generated between the plural transport layers.

9. The system of claim **1**, wherein the support frame includes a top body, a bottom body, and one or more pillars that separate the top body from the bottom body.

10. The system of claim **9**, further comprising:

a floating element configured to float the plural transport layers on the saline feed so that a bottom part of the plural transport layers is fully located within the saline feed.

11. The system of claim **1**, wherein the plural transport layers include plural microchannels that promote capillarity, and the plural microchannels act as a conduit for salt backflow from the solar absorber layer toward a bottom part of the plural transport layers.

12. The system of claim **1**, wherein the plural transport layers are configured to generate vapors and a condensate of these vapors is generated using only solar light.

13. A solar-driven atmospheric water extraction system comprising:

a support frame;

a mass and heat transport component supported by the support frame, the mass and heat transport component having plural transport layers;

a solar absorber layer located on top of the transport layers; and

a sorption system in which a bottom of the plural transport layers is located,

wherein the plural transport layers include plural microchannels that support capillarity, promote a flow of atmospheric water toward the solar absorber layer, generate vapors due to heat generated by the solar absorber layer, and absorb atmospheric water; and

wherein the solar absorber layer is formed directly on top of the plural transport layers.

14. The system of claim **13**, wherein the sorption system includes a sorption fluid that travels through the plural transport layers toward the solar absorber layer, and the sorption fluid absorbs the atmospheric water from atmosphere and carries the atmospheric water through the plural transport layers toward the solar absorber layer.

15. The system of claim **13**, wherein the plural transport layers are parallel to each other, and the plural transport layers extend along a direction that is perpendicular to the solar absorber layer.

16. The system of claim **13**, wherein the plural transport layers include 32 layers and the plural transport layers have a top water evaporation zone that is about 3 cm tall, for generating water vapor, and a bottom atmospheric water capture zone that is about 5 cm tall, for capturing the atmospheric water.

17. The system of claim **13**, wherein heat generated by the solar absorber layer heats the sorption liquid and the atmospheric water, which flow through microchannels of the plural transport layers, and generates vapors.

18. The system of claim **17**, further comprising:

a cover located over a top portion of the plural transport layers, the cover being configured to condensate the vapors generated between the plural transport layers.

19. The system of claim **13**, wherein the support frame includes a top body, a bottom body, and one or more pillars that separate the top body from the bottom body.

20. The system of claim **13**, wherein the plural microchannels act as a conduit for sorption particles backflow, from the solar absorber layer toward a tank of the sorption system.

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